A Bribery-Resistant Group-Strategyproof Oracle

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Abstract

The integrity of many high-value protocols, including prediction markets, rests upon the accuracy and reliability of information they draw from oracles about the outside world. We propose Migration-Based Universe Forking with Supply Restoration (MBUFSR) as a form of decentralized oracle significantly more resistant to attack than all current offerings. In game-theory terms, it establishes truth-telling as a strictly dominant strategy equilibrium, offering a more robust defense against attacks by using forking to leverage an environment in which a network's value depends on maintaining a sustained truthful equilibrium. To assess the relative strength of MBUFSR's security model, we group the world of decentralized oracles into four classes and formally derive robustness formulas that enable comparing attack costs under any real-world scenario.

Our protocol empowers anyone to pay a fee to verify any real-world fact while outperforming existing oracles at resisting external influences. It delivers unprecedented security against attacks on its integrity: the cost to cause a fraudulent outcome is equal to twice the cost of buying and then destroying 50% of the supply. When participants act rationally, what is considered truth is not decided by token holders, governance, multisigs, or anyone else - it emerges organically from market forces. This approach offers a trustless pathway to efficiently distill humanity's collective knowledge of truth more securely than alternatives, better supporting dependent systems including, but not limited to, fully decentralized prediction markets.

1 Background

Humanity has long grappled with the challenge of discerning true knowledge amid a sea of dispersed, partial, and often conflicting beliefs, as no centralized authority can reliably aggregate the totality of human understanding. This quest for consensus has historically relied on institutions, experts, and communal voting processes. Yet, each approach has fallen short in achieving impartiality and resilience against manipulation. The difficulty centralized authorities have when trying to approximate truth arises from a fundamental challenge with the aggregation of knowledge. As Friedrich Hayek described in *The Constitution of Liberty*,

> The sum of the knowledge of all the individuals exists nowhere as an integrated whole. The great problem is how we can all profit from this knowledge, which exists only dispersed as the separate, partial, and sometimes conflicting beliefs. — Friedrich Hayek[4]

This insight was framed within the context of the knowledge of value, arguing that the optimal allocation of resources is inherently unknowable to any single entity. Although his focus was economic, the underlying point reveals a more general truth about the nature of knowledge aggregation: no centralized authority can rival the ability of the open market to capture the dispersed and partial understandings held across society.

Our oracle model extends this principle beyond economics. Implicit in its design is the idea that truth is best discovered as an emergent property of decentralized market processes. Although humanity's collective knowledge does not exist as an integrated whole, as Hayek observed, it can be approached through mechanisms where self-interested actions converge toward a game-theoretic equilibrium that reflects our best collective understanding of reality.

These mechanisms must incorporate an enforcement component to function effectively. Achieving unbiased enforcement demands a credibly neutral arbiter capable of upholding contractual agreements without susceptibility to manipulation. The advent of blockchain technology has enabled consensus mechanisms that

provide precisely such a medium, avoiding centralized points of failure and enabling a more robust pathway to discovering truth. As blockchains are the only known method to achieve such an enforcement component, they will occasionally be referenced during this paper.

2 Protocol Summary

We introduce an open, decentralized, trustless oracle that publishes truth for permissionless use by third parties. Secured purely by economic incentives and market forces, it is designed to serve as an infrastructure layer for prediction markets, DeFi systems and other protocols that wish to outsource resolution to a superior, more reliable oracle.

The purpose of the oracle is to answer any question about what is true. Anyone may pay a fee and provide valid parameters to activate the oracle, which autonomously resolves any multiple-choice question without any centralized backstops or vetoes. When a Query is ready for resolution, the fee bootstraps the labour cost for a reporter (anyone wanting a portion of the fee) to optimistically propose a tentative outcome by placing a bond equal to the fee. If the proposed tentative outcome is not aligned to truth, anyone can profitably dispute it, ensuring that even with a successful attack, an attacker always takes on a loss within the protocol.

2.1 Escalation Game

When reporters are disputed, a game begins that we call the Escalation Game. An initial report starts timed rounds. Anyone can dispute the tentative outcome by posting a bond twice the previous bond, replacing the tentative outcome for the next round. If a tentative outcome goes undisputed for its round, the oracle resolves that outcome as truth and transfers almost all of the losers' bonds to winners. This profit motive aligns economic incentives: an attacker must outspend the profit-seeking market to succeed, making honest defense rational, profitable, and easily crowdsourceable. Most queries are expected to resolve during the escalation game, but a sufficiently motivated attacker could still outspend defenders, for this we use an algorithmic fork as a backstop.

2.2 Algorithmic Fork

If the escalation reaches a large enough size, the system codifies the social defense by triggering an algorithmic fork. This fork duplicates all existing queries into new universes, one for each forking Query outcome. A two week, one-way migration window then forces the holders of the revenue-correlated asset (REP) to choose which universe to follow, with legacy REP being deprecated after migration. Rational REP holders are expected to migrate only to the honest universe, as nobody would use an attacker-controlled universe, making it economically worthless.

After the migration, the REP supply is restored by minting equal to the supply decrease of the most popular fork and then sold by auction. Given that the attacker must maintain majority control, this supply restoration forces the attacker to purchase another 50% of the supply and migrate it to false, further increasing the cost of attack.

To misresolve a Query, an attacker must ensure that their fraudulent fork wins the forking game, but doing so under repeated restoration and migration requires them to buy and burn 50% of the supply twice – first to carry the migration, then again to restore supply through a mint-and-auction. This yields an effective attack cost exceeding 100 % of the oracle's market cap, making dishonesty economically expensive.

After the auction, the oracle then finalizes resolution to the universe with the most migrated tokens. In practice, even if the attacker wins and becomes the majority, economic value will still only reside in the honest universe.

3 Survey Summary

This paper additionally provides an unprecedented overview of decentralized oracles, breaking them into 4 classes while formally deriving and comparing their attack costs under real world scenarios lacking complete

rationality and information.

- We find that the most expensive to attack oracle design is the one introduced in this paper, Migration-Based Universe-Forking with Supply Restoration (MBUFSR), which has a likely attack cost that is 134% of the fully diluted valuation of the voting asset (FDV).
- In second place is Migration-Based Universe-Forking (MBUF) used by Augur V2 at 92% of FDV.
- In third place is MBUF without mandatory migration used by Kleros V2 and Eigenlayer at 83%.
- In fourth place is Schelling-point governance voting with social recovery at 73% which requires the developers in advance to make a credible threat to: manually fork, revert all holdings back in time to the fraudulent vote, and destroy shares that voted fraudulently.
- In fifth place is Price-Based Universe-Forking at 58%.
- In last place is normal Schelling-point governance voting as used by UMA, Kleros V1, Tellor, Band, and Entangle; which in a theoretical environment where hedging is free has zero attack cost, but in practice has an attack cost that we estimate is 26% of FDV.

While the precise attack cost values presented are contingent upon the underlying assumptions and predictions employed in our analysis, readers are encouraged to substitute their own parameter estimates into the derived formulas to explore alternative scenarios. These assumptions can be found in Appendix.6 For example, we assume buying 50% of the supply to cost far more than 50% of the FDV. Nonetheless, we show in our analysis that altering this relative ranking of oracle designs would necessitate the expectation of real-world conditions that may be deemed improbable or unrealistic.

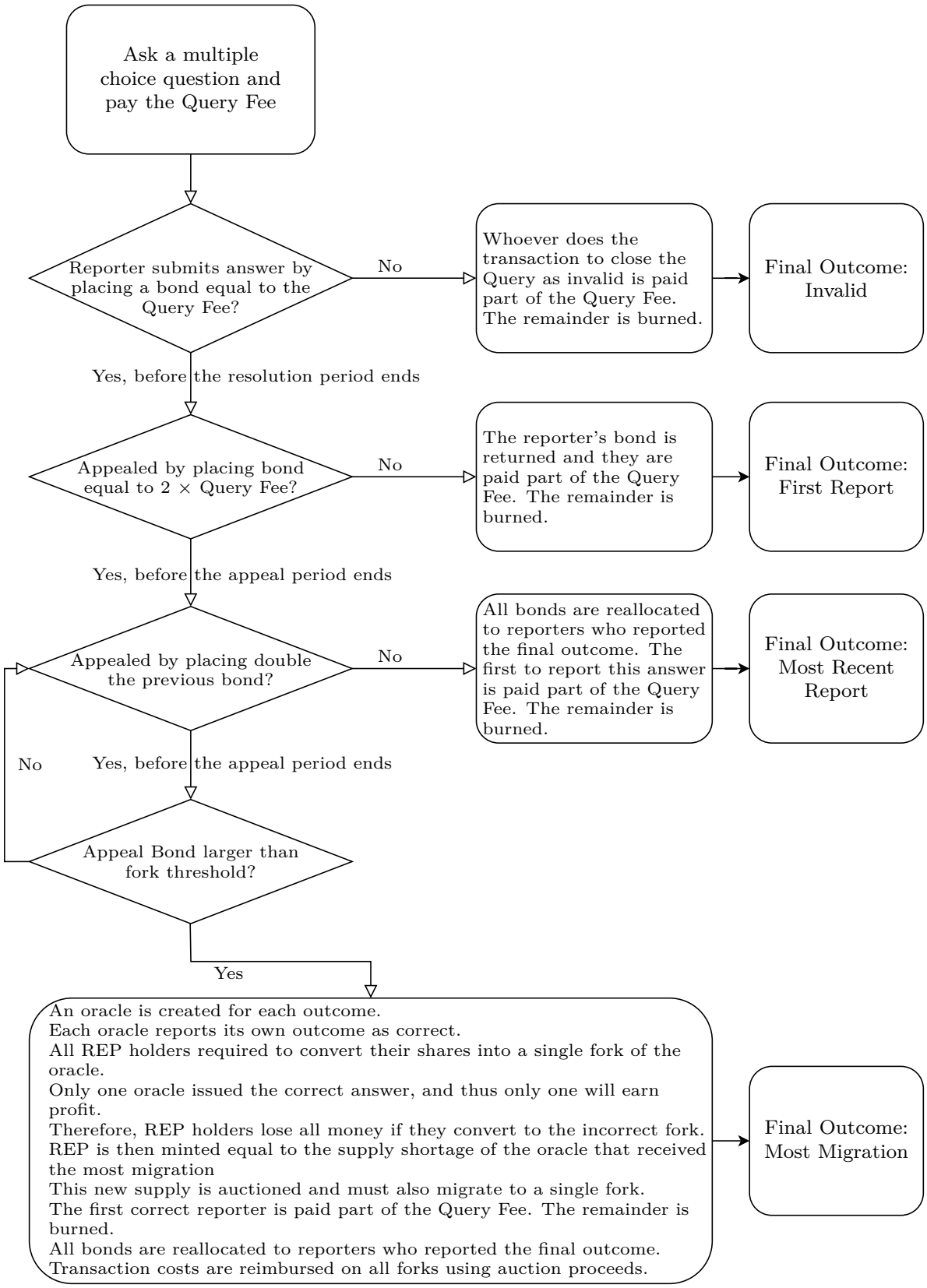
4 Specification

Compared to the industry oracle mechanisms surveyed later in the paper, we identify migration-based universe forking with supply restoration as the oracle mechanism offering the highest robustness against external incentives while maintaining alignment to ground truth, we now turn to a detailed specification of the protocol. In what follows, we describe an implementation of MBUFSR that incorporates dynamic parameters.

4.0.1 Terminology

The protocol's primary objective is to answer Queries. A Query is defined as a tuple consisting of a natural-language question (e.g., "What was the outcome of the 2024 U.S. Presidential Election?") and a finite list of mutually exclusive and exhaustive possible outcomes. This list of outcomes includes "Invalid" to accommodate malformed or unresolvable queries. Anybody can become a Query Creator by submitting a valid set of parameters accompanied by a Query Fee. Valid queries are eventually converted into Final Outcomes by Reporters who place bonds on tentative outcomes via a multi-layered escalation procedure. Throughout this specification, we adopt the term "REP" as a concise designation for the revenue-correlated asset, the valuation of which is intrinsically bound to the oracle's persistent adherence to truthful answers. REP is used for all value transfers in the protocol including reporting, disputing, Query fees, reporter pay, and protocol earnings.

4.0.2 Process Summary



The process by which the oracle answers a question. A more detailed diagram can be found in Appendix.

4.1 Escalation Game

The typical oracle process involves someone reporting ground truth where no one disagrees, causing the initial report to become the Final Outcome. This is typical because reporting requires placing a bond that is lost if the initial report is wrong. What happens in the event of a disagreement is called the "Escalation Game". This is analogous to lower courts, from which disputes can be appealed through increasing appeal bonds to a supreme court where the supreme court in this analogy is the Forking Game (described later).

The escalation game consists of a sequential staking process in which participants bet REP on their belief that a specific outcome will ultimately prevail as the Final Outcome for the Query. The first stake has a minimum size equal to the Query Fee. Each subsequent stake must be placed on a different outcome and twice the amount of the previous stake. If no stakes are placed for an amount of time referred to as the Appeal Period, the outcome endorsed by the most recent stake becomes the Final Outcome. If the required stake would be greater than a threshold termed Fork Threshold, the required stake is changed to this amount and the game changes to the forking game when it is filled (described in Specification: Forking Game).

The purpose of this escalation game is to minimize the capital barrier to entry for reporting while preserving incentive alignment and equilibrium robustness.

4.2 Forking Game

4.2.1 Fork Threshold

The forking game begins when a bond equal to the Fork Threshold is staked on an outcome. The Fork Threshold parameter is a percent of the entire supply of REP. If set too low, byzantine attacks that frivolously fork REP become too affordable; if set too high, honest parties may not be able to gather enough money to contest attackers. Setting this parameter anywhere from 0.5% to 4% of the supply appears reasonable.

4.2.2 Migration

In the first stage of the forking game every REP holder chooses between inaction and a list of all outcomes for the disputed Query. If they choose inaction, their REP is effectively destroyed because it will never again earn profit. If they choose an outcome, their REP is converted to the fork of REP associated with that outcome. This conversion is referred to as migration.

All REP that is staked in an escalation game that is not the subject of the fork is refunded and also required to migrate. The reporting process for all existing queries starts over as a result and can not resume until after the fork. The only REP exempt from this migration requirement are Query Fees and REP staked in the escalation game that is the subject of the fork. This REP instead splits into every fork. This exemption exists so that reporter pay and the opposing stakes in the escalation game can be paid to the winner in each universe. Note that Queries can still be created during the forking game if they are manually directed to one of the children of the parent fork.

This migration results in a complete loss of the REP investment unless the holder chooses the valuable fork and they experience no loss otherwise. The complete-loss payoff assumes that only one fork will have customers. The no-loss payoff assumes that the REP price will not decrease as a result of the fork.

We suggest a time-frame of weeks for the migration phase to maximize anticipated participation and consequently the security margin.

4.2.3 Supply Restoration

In the second stage of the forking game, the supply of the fork that received the most migration is restored. Specifically, a quantity of shares of the original protocol equal to the amount not migrated to the most popular fork is created and sold by auction. Then these shares are required to migrate to a fork. If the supply is still not restored due to migration to multiple forks, the supply restoration is repeated and followed by another migration. These two stages can be repeated an unlimited number of times as set by a parameter, however, each repeat increases the security margin by half as much as the previous.

A key insight enabling this supply restoration without diluting shareholders is that an attacker's migration of shares to a fraudulent fork reduces the effective supply for remaining holders, increasing their proportional ownership constituting a value transfer from the attacker to them. This leaves an opportunity for the protocol to restore the supply to its original level by issuing new shares, compelling the attacker to acquire and redirect additional capital to the false branch to maintain their majority ownership of the supply. Shareholders are not diluted because the supply increase exactly equals the supply decrease caused by the attacker.

We suggest a time-frame of weeks for the supply restoration phase to maximize the amount of capital available to participate in the auction and consequently the security margin.

4.2.4 Winning the Forking Game

The fork that received the most total migration and that fork's associated outcome are the winners of the forking game.

4.3 Final Outcome

A Final Outcome is established in one of three ways: First, if a Query receives a Report that is not appealed, the outcome endorsed by that Report becomes the Final Outcome. Second, if a Query does not receive a Report within 3 days it resolves as invalid. Third, if a fork occurs the Final Outcome differs among the copies of the protocol; the original protocol resolves to the winner of the forking game while each fork implements their own outcome as the Final Outcome. The Final Outcome from the original protocol is the "official" answer from the oracle that is typically used by services that rely on the oracle.

4.3.1 Final Outcome Payoffs without Forking

The typical Final Outcome scenario is when REP is staked on an outcome without being appealed. In this case the outcome most recently staked on is the winner. 80% of REP staked on other outcomes is transferred to those who staked on the winning outcome proportional to stake; the 20% remainder is a fee that is destroyed as profit. This 20% fee on escalation game winnings prevents byzantine attacks that delay resolution at no cost by appealing one's own reports. The first party to stake on the winning outcome is additionally awarded the Reporter Pay.

4.3.2 Final Outcome Payoffs after Forking

Each fork of the protocol treats its associated outcome as the Final Outcome. As such, the payoffs after forking are the same as if a fork had not occurred with each fork carrying out the steps from the previous subsection. The only difference is that the payoffs are denominated in forked REP. These payoffs are worthless for any REP holder that migrated to a false universe; even if an attacker causes the false universe to win.

4.3.3 Final Outcome in the Original Protocol

The original pre-fork version of the protocol technically ceases to operate after a fork; however, it does not appear that way. For backwards compatibility and continuity of prediction markets and other use cases that interact with the original protocol, it begins forwarding all interactions to the fork that won the forking game. The original protocol appears to work just the same but what is actually happening is that interactions with the protocol are forwarded to the winning fork.

4.3.4 Minting Auction Proceeds

As previously mentioned, during the supply restoration phase, additional REP is minted and sold by the original protocol. What remains unaddressed is determining a destination for the proceeds of this auction. If the proceeds are given to the winning fork, it increases the payoff for the attacker by double what the supply restoration made them spend, overall reducing the security margin. If the proceeds are distributed proportional to migration the same issue occurs except half as much. If the proceeds are destroyed then an opportunity to increase the market cap of the ground truth fork is wasted as well as the wealth itself.

Ideally, the proceeds would be delivered exclusively to the fork that represents ground truth or at least divided proportionally to legitimate revenue. The former is not possible because the protocol does not know which fork is ground truth after an attack since the attacker controls this determination. However, fortunately, the latter option of dispersing proportional to legitimate use is possible. To accomplish this, every fork becomes eligible to receive funds in an amount less than observed economic frictions, such as transaction and opportunity costs. For example, 90% of all blockchain transaction costs can be reimbursed using the auction proceeds until the funds are exhausted. With this reimbursement rule it is only rational to collect the auction proceeds if the blockchain transaction cost would have been spent even without the reimbursement; in other words it is only spent on legitimate use within each fork, which will only occur on non-fraudulent forks. This rule is enough to cause the proceeds to increase the market capitalization of the non-fraudulent fork, and prevent the attacker from profiting from the proceeds.

However, these proceeds are expected to be many times larger than the total blockchain transaction costs of participants and can take years to exhaust the fund. Thus, the subsidy must be time discounted when estimating its augmentation of REP valuation. This time discounting can be further mitigated by reimbursing additional forms of economic friction such as opportunity cost. When the subsidy is used in this way to increase the valuation it also increases the amount the attacker must spend, and consequently the security margin.

4.3.5 Process Summary

At a high level, the protocol processes each Query through the following stages: (1) Query creation and fee payment, (2) initial reporting of a tentative outcome, (3) an appeal period where anyone can claim the initial reporter is incorrect through an escalation game of increasing bonds, (4) forking initiation if appeals reach a predefined threshold, comprising a migration phase and a supply restoration phase, (5) outcome finalization, and (6) distribution of rewards and penalties to reporters. We describe each stage in detail below.

4.4 Fees and Bonds

4.4.1 Query Creation

Any participant, referred to as a Query Creator, may submit a Query to the oracle.

4.4.2 Query Fee

When a Query is created, the Query Fee is paid entirely in REP and is split into two payments that serve two purposes. First, compensating reporters for their labor through a payment termed Reporter Pay. Second, increasing the security margin by burning a portion as profit. The protocol is designed to minimize the reporter fee and maximize profit.

4.4.3 Reporter Pay

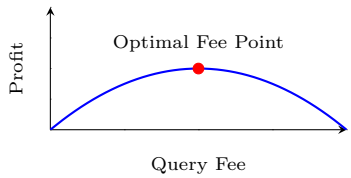
This payment compensates the first participant to report an outcome that becomes the Final Outcome. To avoid overpaying, the protocol uses a clock auction denominated in REP from the Query Fee. When a Query is posted, the reporter reward starts at 0 REP. Over time, the offered reporter reward linearly increases the portion of the Query Fee offered, reaching the entire Query Fee a configurable amount of time later. This creates competition among reporters to accept the lowest possible REP payment, ensuring that reporting occurs at the minimum cost necessary. To encourage faster reporting and reporting before expiration, anyone may optionally add a tip at any time that augments the Reporter Pay.

4.4.4 Profit

Any portion of the Query Fee not paid to the reporter is burned. Burning REP reduces total supply, increasing the value of remaining REP. The expectation of continued supply reduction gives REP market value. This market value is critical for security because the cost to attack the oracle scales with market capitalization. Therefore, an optimal Query Fee is one that maximizes the FDV by maximizing monthly profit.

4.4.5 Query Fee Controller

To achieve this maximization, the fee is governed by a simple feedback controller that dynamically adjusts the Query Fee based on the observed outcome. Specifically, the controller monitors the expected quantity of REP destroyed each month and follows a rule: if a fee change results in more REP destroyed, the fee is shifted further in that same direction; otherwise, it is reversed. This creates a hill-climbing process that incrementally converges towards the optimal fee level at a rate of 10% per month. Equilibrium is reached when the controller alternates between increase and decrease, indicating that any deviation from the current fee reduces overall profit as quoted in REP. Roughly, this target is reached when a 1% increase to the fee would cause a 1% decrease in the quantity of Queries created.



Note that the profit is only realized once a Query resolves which could take longer than a month; this means that if the controller is to respond in time it must estimate the profit that will be generated at the time of Query creation when only the revenue is known. This is accomplished by multiplying the Query Fee paid times the portion that historically has been profit.

4.4.6 Query Fee Coefficient

This controller has a notable limitation: it operates solely on the quantity of REP burned and remains agnostic to fluctuations in REP's market price. As a result, it responds slowly to significant price shifts, potentially delaying convergence to the new optimal fee. For example, if REP's price suddenly halves the controller would require 7 months of consecutive 10% price increases to return the Query Fee to optimal. One potential solution is to directly measure REP's price, but this introduces risks: the relevant metric is REP's purchasing power (a conceptual measure not tied to any single asset), and relying on a hardcoded quote currency like USD could fail dramatically due to events such as hyperinflation or otherwise decoupling from purchasing power thereby introducing a point of eventual failure. A more resilient approach infers price changes indirectly through Query demand patterns. A sharp drop in Query volume suggests overpricing, while a surge indicates underpricing. This data can be used to apply a stabilizing coefficient to the controller. This coefficient adjusts the fee to mitigate dramatic changes in the Query creation rate enabling faster adaptation without external dependencies. Essentially, the goal of this coefficient is to ensure Queries get created at a rate similar to the historical rate, thereby resolving any issues that could be caused by dramatic changes in REP price.

The coefficient is computed by first calculating a ratio R that compares recent Query creation volume to a historical baseline. Specifically, it measures the number of queries in the most recent 3 days against the preceding 60 days, scaled by a factor of 20 for normalization:

$$\text{ratio} = 20\times(\text{queriesCreated.previous3days})/(\text{queriesCreated.previous63Days}-\text{queriesCreated.previous3days})$$

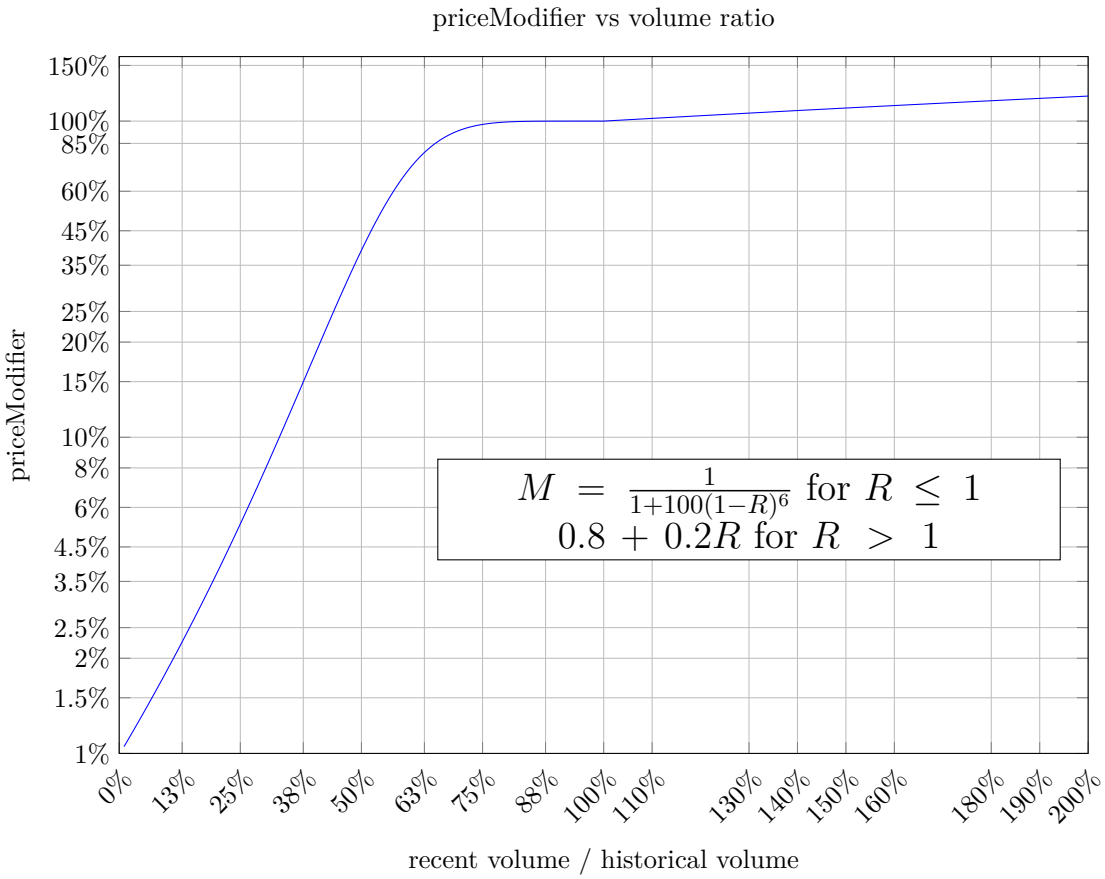
A ratio below 1 indicates lower-than-expected volume, signaling a need to reduce the fee while a ratio above 1 occurs when volume increases.

The 3-day recent interval balances quick detection of demand drops (e.g., from overpricing) with reduced fee volatility, ensuring corrections reach full strength within 3 days. The 60-day historical interval mitigates overreactions to natural growth while avoiding excessive fluctuations that could harm the user experience through unpredictable timing requirements.

This ratio is then transformed into an asymmetric price modifier M to prioritize rapid responses to shortages over surges, since overcharging (low ratios) causes greater harm to utility and revenue than undercharging (high ratios).

For $R\leq1$, we apply a non-linear reduction that is only reactive to significant declines, and for $R>1$, a linear increase is applied:

$$\text{priceModifier} = \begin{cases} \frac{1}{1+100(1-\text{ratio})^6} & \text{if ratio} \leq 1 \\ 0.8 + 0.2 \times \text{ratio} & \text{if ratio} > 1 \end{cases}$$



For instance, if the volume of Queries created in the last 3 days is double what was expected, the Query Fee will be increased by 20% because a 200% ratio maps to 120% price modifier. If the volume is 30% less than expected, the formula largely ignores it and only applies a 7% discount to the Query Fee. After this point, the discount quickly becomes dramatic. A 50% decline in volume leads to a 61% discount for Query Creators, while a 67% decline results in a 90% discount. Further declines rapidly approach the maximum discount of 99%.

The final Protocol Query Fee is the controller's base output multiplied by M.

4.4.7 Query Tokens

Creating a Query normally requires providing a question and paying the current Query Fee. However, some users such as institutions may require predictable costs and bulk purchases of Queries before knowing what questions they will ask. To facilitate broader adoption, the protocol supports tokenizing the right to create a Query via a mechanism called Query Tokens. A Query Token is created by paying the current Query Fee plus a fixed 10% premium, all denominated in REP. This payment does not go directly to the oracle protocol. Instead, the REP is deposited into a separate, trustless, and immutable Query Token protocol. This separation enables prepaid Queries while minimally changing the core protocol. The pooled REP collectively backs all outstanding Query Tokens, with the average REP per token (total pooled REP divided by the number of Query Tokens) representing the fee that will be paid to the oracle when a token is redeemed. Allowing the Query Token contract to pay this different price is the only change made to the core protocol to facilitate Query Tokens. To redeem a Query Token, the holder can destroy it. Upon redemption, the Query Token protocol automatically creates a normal Query in the oracle protocol on the holder's behalf

and transfers REP from its pooled reserve to cover the Query Fee. From the oracle's viewpoint, this is indistinguishable from a direct Query submission, except that a different price is paid. This mechanism provides fee certainty: users pay a known premium upfront to hedge against potential increases. This fixed 10% premium compensates the protocol for bearing the risk of reduced profit in the event the Query Fee would later increase. It is deliberately set equal to the Query Fee controller's maximum monthly adjustment rate, which discourages speculative minting solely to arbitrage anticipated fee hikes. In the event of a fork, the Query Token protocol forks along with the oracle. Query Tokens must migrate to a single fork during the forking period. The token holders declare which fork to migrate to, and the Query Token protocol migrates the associated REP on their behalf. Any unmigrated Query Tokens are destroyed along with the associated REP.

4.5 Example

To summarize the specification so far, the following is an example of the oracle process that is typical except for a decline in volume. The Query Fee controller hill climbs-finding the Query Fee that maximizes profit is 17 shares of REP. By chance or perhaps because the REP price increased, no Queries have been created for the last 24 hours despite being previously constant. As a consequence the Query creation volume over the last 3 days is $\frac{2}{3}$ of the target, so the price modifier is 88%, yielding a final Query Fee of 15 REP. The price modifier would decrease to only 10% over the next 24 hours if Queries continue to not be created, but a market creator thinks of a question that will generate a lot of trading volume if the general public was given the ability to bet on the outcome. The creator expects a significant profit from fees on this trading volume and so they pay the 15 REP Query Fee. They are happy with receiving resolution within 3 days so they do not include a tip. The Query Creator specifies possible outcomes 'Yes' and 'No'. The protocol automatically adds 'Invalid' as an outcome. Some time passes where prediction market participants pay the creator fees as they bet on the outcome, eventually generating profit for the creator once their revenue exceeds 15 REP. Since the Query Creator did not include a tip, the protocol does not yet offer payment, however anyone is welcome to report without pay; those betting on the outcome will sometimes do so. The protocol starts offering an increasing portion of the Query Fee to the first person to report the winning outcome. 12 hours later the protocol is offering $\frac{1}{6}$th of the Query Fee which is enough to attract a Reporter. A Reporter stakes 15 REP on 'Yes' because their bond is required to equal the Query Fee and because reporting any other outcome would result in the loss of their bond. One day passes without anyone posting the 30 REP bond required to appeal. 'Yes' becomes the final outcome. The 15 REP bond is returned to the Reporter along with a 2.5 REP payment. The remaining $\frac{5}{6}$ths of the Query Fee is destroyed, effectively delivering a dividend worth 12.5 REP to shareholders.

4.6 Attacking the Forking Game

To enforce a fraudulent outcome as the Final Outcome for a Query, an attacker must ensure that their preferred (fraudulent) fork prevails in the forking game. Achieving this dominance requires a multi-step process: first, acquiring at least half of the amount of REP that will participate in a fork; second, escalating a Query through appeals until the forking threshold is reached; third, migrating all acquired REP to the fraudulent fork, rendering it worthless; fourth, purchasing half of the newly minted REP during the supply restoration phases; and fifth, migrating this additional REP to the fraudulent fork as well. These actions will cause the fraudulent fork to win the forking game, but doing so is very expensive as it requires buying and then effectively destroying half the REP supply twice over.

To realize a net profit despite this large expense, the attacker must simultaneously accumulate a substantial position in the unlikely outcomes of financial contracts that depend on the oracle for resolution, such that the redirected payouts from the fraudulent Final Outcome outweigh the in-protocol losses incurred.

4.6.1 External Incentive to Attack the Forking Game

Within a migration-based universe forking oracle, rational participants always receive the greatest payoff from migrating their holdings to the fork that they believe best matches ground truth. This applies even to potential attackers, who will inevitably lose money within the protocol by supporting a fraudulent outcome.

However, an attack can make sense if external incentives come into play. In other words, if the gains from fraud outside the oracle exceed the losses inside it, fraud becomes profitable.

This incentive to attack an oracle extends beyond the immediate fraud enabled by altering a single Query's outcome. It more broadly arises in any scenario where third-party financial contracts blindly adhere to the winning fork's outcome in the forking game. If an attacker ensures victory for a fraudulent fork in the forking game, they gain the ability to dictate resolutions for all pre-existing Queries, enabling exploitation of all financial contracts that are immutably dependent on the oracle.

The external incentive for such an attack can be quantified as the net value extractable from financial contracts contingent on the oracle's outputs. This is calculated as the total capital at stake in these contracts (representing the wealth transferable upon resolution) minus the cost of acquiring the cheaper side of the positions (those aligned with the less probable outcomes). In prediction markets, for instance, this equates to the total open interest less the expense of purchasing all shares of the least likely outcomes.

In practice, however, attackers cannot acquire literally all unlikely shares and the portion they do acquire will cost a premium due to price impact. Three primary factors reduce the fraction of total open interest that constitutes the effective incentive for fraud: first, the least likely outcome for each market has a starting price the attacker must pay ranging from 0% to 50% of the winning payout; second, large-scale purchases typically incur premiums; third, acquiring literally all shares is likely prevented because a portion of holders may refuse to sell even at a premium. Consequently, a reasonable estimate of the effective motive for the attacker as a fraction of total open interest ranges from 60% to 90%. For example, if the average least likely outcome has odds of 20%, and 80% of outcome holders are willing to sell for an average price of 0.25, then the motive to attack would be 60% (0.8*0.75) of vulnerable open interest.

4.6.2 Preventing Attacks

Attacks on the oracle become rational only when the external incentives for fraud surpass the attack cost which is proportional to the market capitalization of REP. These external incentives are collectively referred to as vulnerable open interest, which is defined as the total value that a would-be attacker can acquire by causing the oracle to resolve in the way most favorable to them.

The protocol thus operates within one of two distinct regimes. In the first, the market capitalization of REP substantially exceeds the product of vulnerable open interest and the security margin, rendering profitable attacks inherently infeasible. In the second, vulnerable open interest rationally attempts to self-limit to ensure attacks remain unprofitable. The following subsections examine the potential for attacks arising from breakdowns in this self-limitation under the latter regime.

4.6.3 Incomplete Information Attacks

When participants have complete information, the vulnerable open interest will not rationally exceed the attack threshold. To illustrate, consider a market participant who is contemplating an increase in vulnerable open interest while at the attack threshold: such an addition would anticipate a negative expected payoff, as the optimal response from a rational attacker would lead to the fraudulent redirection of contract payouts. In essence, it is irrational for participants to expand vulnerable open interest beyond the point where an attack becomes viable, implying that attacks should never occur.

However, if the 'complete information' assumption is dropped then the vulnerable open interest will only rationally exceed the threshold under a simultaneous combination of six conditions:

1. A substantial portion of vulnerable open interest exists that is unknown to most participants. Given the inherent transparency of blockchain-based systems, this hidden interest must reside either in centralized platforms that resolve using the oracle even after evident fraud or in encrypted smart contracts that obscure transaction values and positions.

2. Participants significantly underestimate the scale of hidden vulnerable open interest.

3. The hidden vulnerable open interest is dispersed across multiple independent parties, each unaware of the others' positions. (A single entity or internally transparent group holding all hidden vulnerable open interest would have complete information and rationally reduce positions preventing profitable attacks.)

4. This hidden vulnerable open interest permits an external attacker to participate by acquiring positions within those contracts.

5. The attacker knows that the hidden vulnerable open interest is significantly greater than the estimate of other participants.

6. The attacker possesses wealth greater than the attack cost plus the cost of purchasing the unlikely side of enough vulnerable financial contracts to recover the loss required to win the forking game.

Collectively, these conditions delineate a scenario in which a large share of vulnerable open interest is fragmented across encrypted financial contracts accessible to anonymous traders, with a malicious actor who is wealthier than the oracle's entire market capitalization and is privy to the scale of hidden vulnerable open interest while the involved parties and public remain oblivious.

This scenario becomes even less likely when there are information tools available. The information required to mitigate this type of attack is an accurate accounting of the visible vulnerable open interest and an estimate of the portion that is hidden yet accessible to an attacker. This objective can be met by a community-maintained dashboard that aggregates all known vulnerable open interest along with estimates of potential hidden amounts, presenting them in a unified view. Moreover, user interfaces for oracle-dependent markets could integrate real-time warnings, alerting traders that current conditions render attacks rational and advising either to abstain from creating new contracts or to withdraw existing positions preemptively.

4.6.4 Friction Attacks

Additionally underpinning the claim that rational behavior precludes attacks is the assumption that vulnerable open interest can contract as quickly as the attack cost; a value proportional to REP's valuation. Theoretically, participants act the moment it becomes rational to do so. In practice, however, friction results in delayed reactions. As a result, changes in REP price can make an attack affordable more quickly than vulnerable open interest can react.

The impact of this scenario is mitigated by real-world factors. If the attacker buys in advance, they suffer the losses of the price crash. If instead an attacker waits until a large price decrease to begin the attack, they will need to acquire half of the REP supply faster than vulnerable open interest can reduce. This massive purchase within a short timeframe may not be possible due to limited liquidity, and even with sufficient liquidity the price impact of such a purchase can easily result in an average entry price greater than before the crash, again rendering the attack unprofitable.

This scenario is also mitigated by risk aversion. Market participants are likely to keep vulnerable open interest somewhat below the threshold, building a margin of safety. This buffer helps protect against sudden price drops in the REP and hidden open interest.

4.6.5 Attack Summary

Under complete rationality, perfect information, and frictionless conditions, an attack is not rationally expected to occur. Even absent these ideals, the probability remains low, and it can be further mitigated via the natural margin of safety from participants' risk aversion, the existence of informational services that estimate vulnerable open interest, and the realistic likelihood that REP's valuation and consequent security margin far exceed vulnerable open interest.

5 Robustness and Alignment in Decentralized Oracles

Oracles deliver reports on objective ground truth in contexts where substantial financial stakes depend on the outcome. Accordingly, these oracles must achieve two principal objectives: alignment with ground truth and resilience against external incentives.

Establishing truth-telling as a strictly dominant strategy equilibrium is straightforward, but ensuring its robustness against external incentives poses greater challenges. Likewise, it is trivial to achieve alignment to a Schelling point, but challenging to align to ground truth.

5.0.1 Defining Ground Truth

Ground truth is traditionally defined as objective reality as perceived by an omniscient observer. In practice, agents can only approximate this truth to varying degrees of accuracy. For brevity, throughout this paper, we will use the term 'ground truth' to refer to the market's best feasible approximation of the underlying truth. This usage is distinct from the game-theoretic notion of 'truth' or 'honesty', which refers to the honest revelation of information by a participant.

5.0.2 Measuring Robustness

The robustness of an honest equilibrium against external incentives can be measured by the cost required to change the equilibrium. Prior works refer to this concept by varying names including "attack cost", "cost of corruption", "economic security", "attack resistance", "collusion resistance", and "manipulation resistance". We refer to this cost as the security margin when it is standardized relative to the fully diluted valuation (FDV) of the voting asset, expressed as a percentage of FDV. For example, if changing the equilibrium to fraud requires an in-protocol loss equal to the entire market cap of the voting asset, the security margin is 100% of FDV. This metric will be applied to evaluate key existing mechanisms, beginning with the foundational Schelling-point voting approach.

5.1 Schelling Point Voting

The first implemented mechanism with an equilibrium of honesty regarding arbitrary real-world events is shareholder voting with a redistribution mechanism that transfers value from minority to majority voters. This method was first proposed by Paul Sztorc's Truthcoin[12] and then two months later by Vitalik Buterin in his Schellingcoin post[1]. In the absence of external incentives such as a prediction market, voters align to a Schelling point where the selected "truth" is the most common prediction of others' anticipated choices. Majority shareholder voting with a financial penalty for voting out of consensus has since become the most popular decentralized oracle mechanism, implemented by nearly all decentralized oracles including UMA, Kleros V1, Tellor, Band, Entangle, and more.

With majority shareholder voting, it is possible to cause a fraudulent outcome by purchasing more than half of the supply for the purpose of malicious voting. This act is known as a 51% attack. An attacker simply has to buy a majority share, vote, and then sell. This trade can come at a loss for the attacker because successful attacks likely reduce the value of the shares. For example, if it cost $100 to acquire 51% of the tokens, after the attack you might only be able to sell the tokens for $80 due to the value drop, making the cost of attack $20. The share value decline from a successful attack can be mitigated or avoided through shorting and derivatives. Theoretically, in a frictionless market with infinite leverage, the security margin of a Schelling-point oracle is 0%. Although an attacker must temporarily control a majority of the token supply, they need not incur losses to do so. The attacker can borrow the required tokens, hedge the full position through short exposure, cast a malicious vote, and unwind both positions at par, resulting in no net loss. However, the security margin rises to around 26% of FDV under our estimation of real-world frictions such as trading costs, the lack of counterparties required to hedge at scale, and an anticipated price decline following an attack.

The security margin of Schelling-point oracles can be significantly increased by a credible threat to manually fork the protocol whereby an identical copy is created except with the attackers shares destroyed. In theory, such a fork will prevent the attacker from recovering the unhedged portion of their attack investment. This act is often referred to as "social recovery". However, this method is only effective to the extent market participants are rational and have complete information. For this threat to be fully effective all offers to buy shares of the oracle must be withdrawn the moment the vote happens. If the market price does not immediately drop to zero, the attacker can still sell the shares they used to vote. In practice, some liquidity will continue to exist after the vote preventing the price from immediately dropping to zero. For this reason shareholder voting protocols often make shares non-transferrable for a short time after a vote to give the market time to respond. When protocols credibly make such a threat the value the attacker can recover by selling for is largely eliminated; in our estimation with some irrationality and hedging, this threat roughly triples the attack cost of Schelling-Point voting to around 73% of FDV. This increase from 26% to 73% is a result of changing the likely input for attackedValuation thereby modifying the attackers share value loss

from 30% to 85%. We predict that social recovery will not cause a 100% loss for an attacker due to hedging and failure of liquidity providers to immediately withdraw.

5.2 Peer Prediction Voting

An issue with Schelling-point voting is the type of 'truth' it aligns to. Voters rationally vote how they think others will vote which includes the perception they have about the bias of others. Schelling-point voters may collectively perceive a typical bias among others that is actually rare, possibly leading to a worse approximation of ground truth. For instance, voters might overestimate the prevalence of an ideology that skews interpretations of facts, amplifying minority biases and deviating further from ground truth.

Peer Prediction Voting addresses this problem by requiring reporters to declare two things, their personal report of the correct oracle outcome, and their prediction of how others will report. Reporters maximize payoffs by: (1) aligning their peer prediction with reporters who have the same personal report, and (2) ensuring their peer prediction accurately reflects the overall distribution of personal reports. Notice that these payoffs include no incentive for one's personal report to match the majority consensus. The majority personal report becomes the oracle's output.

Analysis within the paper that introduced the concept[10] and the improved version[5] both conclude that with optimal parameters, truthful reporting of personal beliefs becomes a strict Bayes-Nash equilibrium. However, there are some limitations. First, this conclusion critically relies on the assumption of a natural correlation between personal beliefs and the predictions of others personal beliefs. Second, this improvement to equilibrium alignment comes at a cost of security margin.

Using parameters from Truthful Decentralized Blockchain Oracles 2021[2], the threshold of voting shares required for a successful attack is reduced to approximately 41% of the total supply, suggesting around 18% lower security margin relative to Schelling-point voting.

5.3 Universe Forking

A third category of mechanisms is known as "universe forking". Protocols that implement this mechanism allow duplication of the protocol and its associated revenue-correlated asset into two variants, differentiated solely by a contested decision. There also exists a mechanism to orient the oracle's outcome toward the fork anticipated by the market to yield the highest future revenue.

This mechanism aligns towards ground truth by leveraging market incentives: the true branch will sustain value as it correctly resolves dependent contracts (e.g., prediction markets), attracting users and liquidity, while the false branch fails economically. As a result, the market is placed in a position where it must approximate ground truth to the best of its ability.

The incentive to report ground truth is backpropagated from this mechanism to reporters making their initial report. In the final step, ground truth is approximated to the best of future customers' ability, directing their revenue. Prior to this, the market forecasts the direction of such revenue flows. As a result, holders of the oracle's revenue-correlated asset maximize their payoffs by selecting the fork they expect the market to endorse. Ultimately, this induces reporters at the outset to rationally report ground truth to the best of their ability, knowing that deviations would trigger profitable escalation to forking.

Regarding robustness, universe forking protocols derive their security either by directly observing which fork attains greater market value or by employing a fork selection rule whose payoffs are correlated with the market valuation of the chosen fork. An attacker aiming to induce a fraudulent outcome in price-based forking must purchase the entire supply of the worthless fork at a price exceeding that of the valuable fork, or in the case of migration-based forking (where the universe with most shares wins), acquire a majority of the valuable shares and convert them into worthless ones.

Universe forking substantially enhances the security margin compared to Schelling-point voting. We estimate a security margin of 134% for migration-based universe forking with supply restoration and 92% without, relative to our prediction of 73% for Schelling-point voting with social recovery and 26% without. These specific estimates are highly sensitive to our underlying predictions, particularly those concerning price elasticity, post-attack devaluation, and the market share of the less popular fork. Nonetheless, we prove in Appendix.7.6 there exists no scenario in which Schelling-point voting without social recovery is more secure than migration-based universe forking.

The concept of universe forking was first proposed by Augur[9] for its enhanced security and superior alignment compared to Schelling-point-based voting mechanisms. Subsequently, this mechanism was implemented in Kleros V2[7] and EigenLayer's intersubjective forking mechanism[13].

5.4 Ecosystem Forking

Ecosystem forking extends the universe forking approach by duplicating not only the oracle protocol but also an associated ecosystem of assets, contracts, and/or protocols that are native to and governed by the oracle system.

In its simplest implementation, ecosystem forking consists only of a forkable prediction market contract in which the oracle's revenue-correlated asset serves as the only asset people use to bet on oracle outcomes, a configuration sometimes termed *Colored Coins*[14]. Upon escalation of a dispute to a fork, market participants receive their payouts in the forked variants of the oracle's revenue asset, creating a situation where an attacker can only capture the fraudulent branch of open interest, which holds no market value. For the other markets that were not the subject of the fork, the market itself is forked and continues as normal in both universes, with participants paid using both revenue assets. Consequently, participants wagering on oracle outcomes achieve immunity to losses resulting from oracle fraud.

This protective property against oracle fraud can be generalized to a broader ecosystem. In its most expansive form, an entire blockchain infrastructure can be engineered to fork automatically in tandem with the oracle. Amoveo proposed the first ecosystem forking oracle[16]; creating a blockchain that replicates itself along with its assets, contracts, and protocols whenever an oracle dispute triggers a fork.

However, ecosystem forking is subject to certain limitations. Although it provides access to betting assets that are impervious to oracle fraud, such assets must originate within the oracle system and possess an inherent ability to automatically fork with the oracle. This stipulation precludes safeguards for external assets, including anything that existed before the oracle's inception. When bets on oracle outcomes are placed using exogenous assets, such as fiat currencies, securities, or cryptocurrencies, the security margin for truthful reporting remains equivalent to that of Price-Based Universe-Forking. This inability to bet using external assets can be substantially alleviated through the deployment of forkable protocols that synthesize derivatives that mirror the payoffs of external assets. However, the forkability mandate implies that these derivatives must be collateralized exclusively by forkable assets, typically rendering shares of the oracle as the only source of backing. Additionally, the relative volatility between these shares and the derivative necessitates overcollateralization and capital inefficiency, ultimately replacing the risk of oracle fraud with the risk of bad debt and the cost of interest payments. The second limitation is that Ecosystem Forking does not publish an outcome in the event of a fork. This prevents use by third parties unless the outcome during a fork is decided by Price-Based forking, which is less secure than Migration-Based forking under scenarios we find likely.

Among the four classes of trustless oracle mechanisms surveyed above, all except Schelling-point voting align equilibria toward ground truth. Ecosystem forking grants immunity to oracle fraud when betting with forkable derivatives or shares of the oracle, but comes with either unwanted volatility or interest payments to hold a derivative that might default. Furthermore, ecosystem forking is less robust at "telling the truth" than Migration-Based Forking. The primary goal of our work is to enable the existence of an autonomous system that publishes ground truth with maximum robustness against fraud. Accordingly, this paper concentrates on enhancing Universe-Forking.

6 Modeling Oracle Security

6.0.1 Methodology

All oracle protocols surveyed in this paper use, as an incentive mechanism, an asset whose value depends on the maintenance of a sustained truthful equilibrium. The revenue-earing shares associated with the oracle have this property and are ubiquitously used for this purpose. As such, the measurement of the robustness of each oracle protocol can be standardized relative to the fully diluted valuation (FDV) of the voting asset. This cost is expressed as a percentage of FDV and referred to as the security margin. For example, if changing the equilibrium to false requires an in-protocol loss equal to the entire market cap of the voting

asset, the security margin is 100% of FDV. Throughout this section, this metric will be applied to evaluate oracle mechanisms.

Many difficult-to-predict factors impact the cost an attacker experiences. These factors include price elasticity, opportunity cost, transaction costs, auction efficiency, the prevalence of irrational inactivity, the market share of the less popular fork (if any) and the decrease in valuation after a successful attack. As such, our methodology is to estimate the security margin across a wide range of plausible conditions. We will present a formula for each mechanism that combines these factors to calculate the security margin, provide conjecture on the range of plausible inputs, and show that Migration-Based Forking with Supply Restoration generally has greater robustness than other oracle mechanisms.

6.0.2 Terminology

As a compromise for clarity, variable factors will be denoted with words rather than single letters. Note that the word 'false' will refer to the fork that the market expects to receive less profit; in cases of legitimate controversy the 'false' fork might not literally be false, but is still more likely to be false. Put another way, 'false' is the fork the market believes is less likely to be true.

REP a revenue-earning share of an oracle

true-REP the most valuable fork of REP absent market manipulation. In other words, the fork of the oracle that future customers most prefer to use.

false-REP the second most valuable fork of REP absent market manipulation.

naturalFDV $\in (0, \infty)$ the fully diluted valuation of REP before the attacker takes any actions. This factor is proportional to attack cost and thus has no impact on security margin.

falseMarketShare $\in [0, 0.5)$ The natural FDV of false-REP as expressed as a portion of the valuation of all forks. For example, if the market expects false-REP to receive 20% of all revenue after a fork, then falseMarketShare is 0.2. Note that as falseMarketShare approaches 0.5, the validity of both oracle outcomes approaches equal, and thus arguably the harm caused by an attack approaches zero.

attackedValuation $\in [0, 1]$ Represents how much the FDV will remain after an attack. Specifically, it is the combined FDV of all forks after the attacker's last action divided by the FDV before the attacker takes any actions.

subsidyDiscount $\in (0, 1]$ The portion of the value of a subsidy that translates to an increase in FDV when subsidy=naturalFDV. For example, if you expect the FDV of a protocol to increase from \$100m to \$150m upon receiving a donation \$100m that will be spent refunding transaction costs, your value for subsidyDiscount would be 0.5 because you expect half of the subsidy to translate to a valuation increase. This subsidy only exists in the MBUFSR protocol, whereby proceeds from the supply restoration are given to shareholders after a fork. The subsidy will not exactly equal the natural FDV, 'subsidyDiscount' is just a calibration point. The actual portion that will contribute to the FDV will depend on the size of the subsidy. Large subsidies take longer to spend, leading to additional time discounting. To account for this, the portion of the applied portion of the subsidy approaches 100% as the subsidy approaches zero, equals subsidyDiscount when subsidy=naturalFDV, and approaches 0% as subsidy approaches infinity. These goals are met by the following formula:

$$\text{valuation increase} = \text{subsidy} \times \frac{\text{subsidyDiscount}}{\text{subsidyDiscount} + (1 - \text{subsidyDiscount}) \times \left(\frac{\text{subsidy}}{\text{naturalFDV}}\right)}.$$

responsiveREP $\in [0, 1]$ The portion of shareholders that will respond to strong incentives. Examples of strong incentives include the total loss of holdings if they fail to act and the market price being well above the price the holder would sell at.

auctionEfficiency $\in (0, 1)$ Percent of the fair price that REP will sell for when time sensitive (e.g. auction)

premium $\in [1, \infty)$ The average cost paid by the attacker to purchase REP before attacking divided by the starting price. This includes price impact, trading fees and opportunity cost.

forkBond $\in [0]$ For simplicity, this modeling does not use a fork bond. Including a larger fork bond has a negligible impact to the security margin and its proper accounting can double the length of formulas.

securityMargin $\in [0, \infty)$ The security margin is standardized as the ratio between the in-protocol losses required for fraud and the oracle FDV before the attacker takes any action:

$$\text{securityMargin} = \frac{\text{attackCost}}{\text{naturalFDV}}$$

6.1 SP Security Margin

As per the derivation of SP in Appendix.2.6, the security margin of a Schelling-point voting protocol (SP) is:

$$\text{securityMargin(SP)} = 0.5 \times \text{responsiveREP} \times \text{premium} \times (1 - \text{attackedValuation})$$

For example, if 90% is responsive enabling the attacker to obtain a majority vote by purchasing 45% of the supply paying an average price 1.9 times the starting price and then recovering 70% of their investment after attacking due to a combination of hedging and/or the market expecting continued demand, then:

$$\text{securityMargin(SP)} = 26\% = 0.5 \times 0.9 \times 1.9 \times (1 - 0.7)$$

6.2 PBUF Security Margin

As per the derivation of PBUF in Appendix.3, the security margin formula for Price-Based Universe-Forking (PBUF) is:

$$\text{securityMargin(PBUF)} = \text{responsiveREP} \times \text{attackedValuation} \times (1 - 2 \times \text{falseMarketShare})$$

For example, when the false market share is 4%, 90% of shareholders are responsive to incentives, and demand across all forks collectively is reduced to 70% due to an attack, then:

$$\text{securityMargin(PBUF)} = 58\% = 0.9 \times 0.7 \times (1 - 2 \times 0.04)$$

6.3 MBUF Security Margin

As per the derivation of MBUF in Appendix.4, the security margin formula for Migration-Based Universe-Forking (MBUF) is:

$$\text{securityMargin(MBUF)} = 0.5 \times \text{premium} - \text{falseMarketShare} \times \text{attackedValuation} \times (1 - 2 \times \text{falseMarketShare})$$

For example, if the attacker pays an average price 1.9 times the starting price, the false market share is 4%, and demand across all forks collectively is reduced to 70% after a successful attack, then:

$$\text{securityMargin(MBUF)} = 92\% = 0.5 \times 1.9 - 0.04 \times 0.7 \times (1 - 2 \times 0.04)$$

Note that this derivation of the MBUF security margin assumes all shareholders are required to migrate to a fork before the oracle outcome is decided. Some MBUF protocols such as Kleros V2 and Eigenlayer's intersubjective oracle do not destroy shares that refuse to migrate. This lack of expected supply reduction reduces the price the attacker must pay to acquire half of the supply. Specifically, the 'unresponsivePriceAdjustment' from the derivation no longer applies causing the overall security margin to be multiplied by the portion that votes (responsiveREP) reducing the securityMargin formula to:

$$\text{securityMargin(MBUF without mandatory migration)} = \text{responsiveREP} \times \text{securityMargin(MBUF)}$$

6.4 MBUFSR Security Margin

As per the derivation of MBUFSR in Appendix.5, the security margin of Migration-Based Universe-Forking with Supply Restoration (MBUFSR) is:

$$\text{securityMargin(MBUFSR)}=$$

$$\begin{aligned}
&(0.5-\text{falseMarketShare})\,\text{responsiveREP}\ \times \text{premium}\\
&\quad+\Big(\text{auctionEfficiency}(1-\tfrac{\text{responsiveREP}}{2})-\text{falseMarketShare}\Big)\\
&\quad\times\left(\frac{-\begin{bmatrix}1-(1-\text{falseMarketShare})\text{attackedValuation}\big(\frac{1}{\text{subsidyDiscount}}-1\big)(2-\text{responsiveREP})\text{auctionEfficiency}\\ -(1-\text{falseMarketShare})(2-\text{responsiveREP})\text{auctionEfficiency}\end{bmatrix}+\text{root}}{2\big(\frac{1}{\text{subsidyDiscount}}-1\big)(2-\text{responsiveREP})\text{auctionEfficiency}}\right),
\end{aligned}$$

where

$$\text{root}=\sqrt{\begin{matrix}\begin{bmatrix}1-(1-\text{falseMarketShare})\text{attackedValuation}\big(\frac{1}{\text{subsidyDiscount}}-1\big)(2-\text{responsiveREP})\text{auctionEfficiency}\\ -(1-\text{falseMarketShare})(2-\text{responsiveREP})\text{auctionEfficiency}\end{bmatrix}^2\\ +4\big(\frac{1}{\text{subsidyDiscount}}-1\big)(2-\text{responsiveREP})\text{auctionEfficiency}(1-\text{falseMarketShare})\text{attackedValuation}\end{matrix}}.$$

For example, when the false market share is 4%, 90% of REP is responsive, the premium is 1.9, the supply restoration auction is 90% efficient, the FDV is increased by 52% of the subsidy value, and revenue collectively drops to 70% of original after a successful attack:

$$\text{securityMargin(MBUFSR)}=134\%=$$

$$\begin{aligned}
&(0.5-0.04)(0.9)(1.9)\\
&\quad+\Big(0.9(1-0.9/2)-0.04\Big)\left(\frac{-\begin{bmatrix}1-(1-0.04)(0.7)\big(\frac{1}{0.52}-1\big)(2-0.9)(0.9)\\ -(1-0.04)(2-0.9)(0.9)\end{bmatrix}+\text{root}}{2\big(\frac{1}{0.52}-1\big)(2-0.9)(0.9)}\right)
\end{aligned}$$

where

$$\text{root}=\sqrt{\begin{matrix}\begin{bmatrix}1-(1-0.04)(0.7)\big(\frac{1}{0.52}-1\big)(2-0.9)(0.9)\\ -(1-0.04)(2-0.9)(0.9)\end{bmatrix}^2\\ +4\big(\frac{1}{0.52}-1\big)(2-0.9)(0.9)(1-0.04)(0.7)\end{matrix}}$$

7 Comparing Oracle Security

7.0.1 Plausible Scenarios

The security margin of each oracle protocol is sensitive to various assumptions regarding the external state. As such, the methodology of this paper is to compare security margins under many conditions including all scenarios, plausible scenarios, and likely scenarios. Rationale justifying these plausible and likely values can found in Appendix.6.

	Possible Range	Plausible Range	Most Likely
attackedValuation	$[0,1]$	$[0.5,1]$	0.7
subsidyDiscount	$(0,1]$	$[0.08,0.94]$	0.52
responsiveREP	$[0,1]$	$[0.7,1]$	0.9
auctionEfficiency	$(0,1)$	$[0.7,0.99]$	0.9
premium	$[1,\infty)$	$[1.07,5.3]$	1.9
falseMarketShare	$[0,0.5)$	$[0,0.15]$	0.04

7.0.2 Method

We compare the security margins of Schelling-Point (SP), Price-Based Universe-Forking (PBUF), Migration-Based Universe-Forking (MBUF), and Migration-Based Universe-Forking with Supply Restoration (MBUFSR). Generally, no protocol is strictly more secure than another, as multiple factors interact to varying degrees, potentially causing one to outperform the others under certain conditions. To intuitively convey the thresholds at which the generally less secure protocol becomes more secure, we identify the most likely scenario where the protocols become equally robust. Here, "most likely" refers to the scenario closest to our established "likely" parameter values, as quantified by mean squared error (MSE). This is achieved through uniform random sampling of 1 million scenarios, filtering to retain only those where the generally less secure protocol is superior, and then selecting from this subset the scenario with the minimal MSE relative to the "likely" values. At the thresholds provided below the two protocols have roughly equal security margin. If a parameter in the provided threshold scenario is far from our likely value it means that parameter contributes strongly to the generally less secure protocol being more robust than the other.

7.1 MBUF vs SP

With our likely values, MBUF costs 3.5 times as much to attack as SP.

MBUF is always more secure than SP except under one scenario where they are equal. This exception scenario is when attackedValuation is 0 and responsiveREP is 1.

7.2 MBUF vs PBUF

With our likely values, MBUF costs 59% more to attack than PBUF.

Under complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), MBUF>PBUF only when premium>2. This is because in this scenario the difference simplifies to 0.5 * premium - 1.

Under possible ranges, the most likely PBUF>MBUF scenario is when attackedValuation>92%, premium>178%, and responsiveREP>98%. In other words, PBUF tends to be more robust in environments where successful attacks do not cause a devaluation, all holders participate, and the price impact of buying half of the supply is less than we expect.

7.3 MBUFSR vs SP

With values we find likely, MBUFSR costs 5.2 times as much to attack as SP.

Under complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), MBUFSR is strictly more secure than SP as shown in Appendix.8.

Under plausible ranges, MBUFSR is strictly more secure than SP as shown in Appendix.9.

Under possible ranges, the most likely scenario where SP>MBUFSR occurs is when falseMarketShare>33%, attackedValuation<53%, and other factors are close to likely but a little in favor of SP. In other words, SP tends to be more robust in environments with extremely contentious issues that result in a massive price crash if the less desired outcome wins.

False market share is the largest factor because under SP, it is never rational to vote for the outcome that expects less revenue, but under universe forking a rational migration is expected proportional to false market share reducing attack costs. A lower attacked valuation additionally contributes because under SP the valuation decrease after an attack is the primary source of security for SP, while for MBUFSR a lower attacked valuation reduces the amount the attacker must spend during the supply restoration.

7.4 MBUFSR vs PBUF

With our likely values, MBUFSR is 2.3 times more expensive to attack than PBUF.

Under complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), the most likely PBUF>MBUFSR scenario is when premium<153%, auctionEfficiency<36%, and subsidyDiscount<43%.

Under plausible ranges, the most likely PBUF>MBUFSR scenario is when premium<130%, falseMarketShare>15%, auctionEfficiency<71%, and subsidyDiscount<26%. In other words, PBUF tends to be more robust when resolving a very contentious outcome where the market does not value a subsidy (airdrop), and the attacker can cheaply buy half of the supply both before the attack and during the supply restoration.

Under possible ranges, the most likely PBUF>MBUFSR scenario is when premium<154%, falseMarketShare>13%, and auctionEfficiency<50%.

7.5 MBUFSR vs MBUF

With our likely values, MBUFSR costs 46% more to attack.

Under complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), MBUFSR is strictly more secure than MBUF as shown in Appendix.10.

Under plausible ranges, the most likely scenario where MBUF>MBUFSR is when falseMarketShare>15%, attackedValuation<62%, and other factors are close to likely but a little in favor of MBUF. In other words, MBUF tends to be more robust in environments with very contentious issues that result in a large price crash if the less desired outcome wins.

Under possible ranges, the most likely scenario where MBUF>MBUFSR is when falseMarketShare>20%, attackedValuation<73%, and other factors are close to likely but a little in favor of MBUF.

8 Conclusion

Many high-value protocols depend on oracles to provide reliable, accurate information about the outside world, but oracles can be vulnerable to attack. The Migration-Based Universe Forking with Supply Restoration (MBUFSR) mechanism introduced in this paper represents a significant advancement in decentralized oracle design and resilience. The leading decentralized oracle designs were compared for the first time under real-world conditions such as price elasticity, hedging, incomplete information, and legitimately contentious issues. In this analysis, MBUFSR was generally shown to be the most secure. This enhanced security is due to the protocol's ability to compel attackers to incur substantial in-protocol losses that are equivalent to acquiring and destroying half of the supply twice over. This mechanism achieves unprecedented robustness against external incentives with an attack cost estimated at 134% of the fully diluted valuation (FDV), surpassing all surveyed alternatives, including Schelling-point voting (26-73% FDV) and other Universe Forking variants (58-92% FDV).

By placing the market in a situation where it must estimate ground truth via resource allocation and leveraging this allocation as an incentive mechanism, Universe-Forking establishes reporting ground truth as a strictly dominant strategy equilibrium. This alignment with ground truth demonstrates the broad scope of Hayek's insight that collective knowledge emerges most accurately from a decentralized market-driven process.

This paper also offers a specification of MBUFSR that is friendly to generalized use across diverse use cases. This is accomplished through mechanisms that automatically maximize profit and minimize costs. It is further enhanced by offering the ability to "lock-in" a current price using Query Tokens, making the protocol more suitable for institutional applications.

Although challenges such as insufficient diversity of perception and potential hidden vulnerable open interest remain, the latter is mitigated through informational tools, participant risk aversion, and the inherent economic barriers to attack the protocol. MBUFSR thus offers a robust, practical framework for decentralized truth discovery, setting a new standard for establishing an unbiased arbiter of truth.

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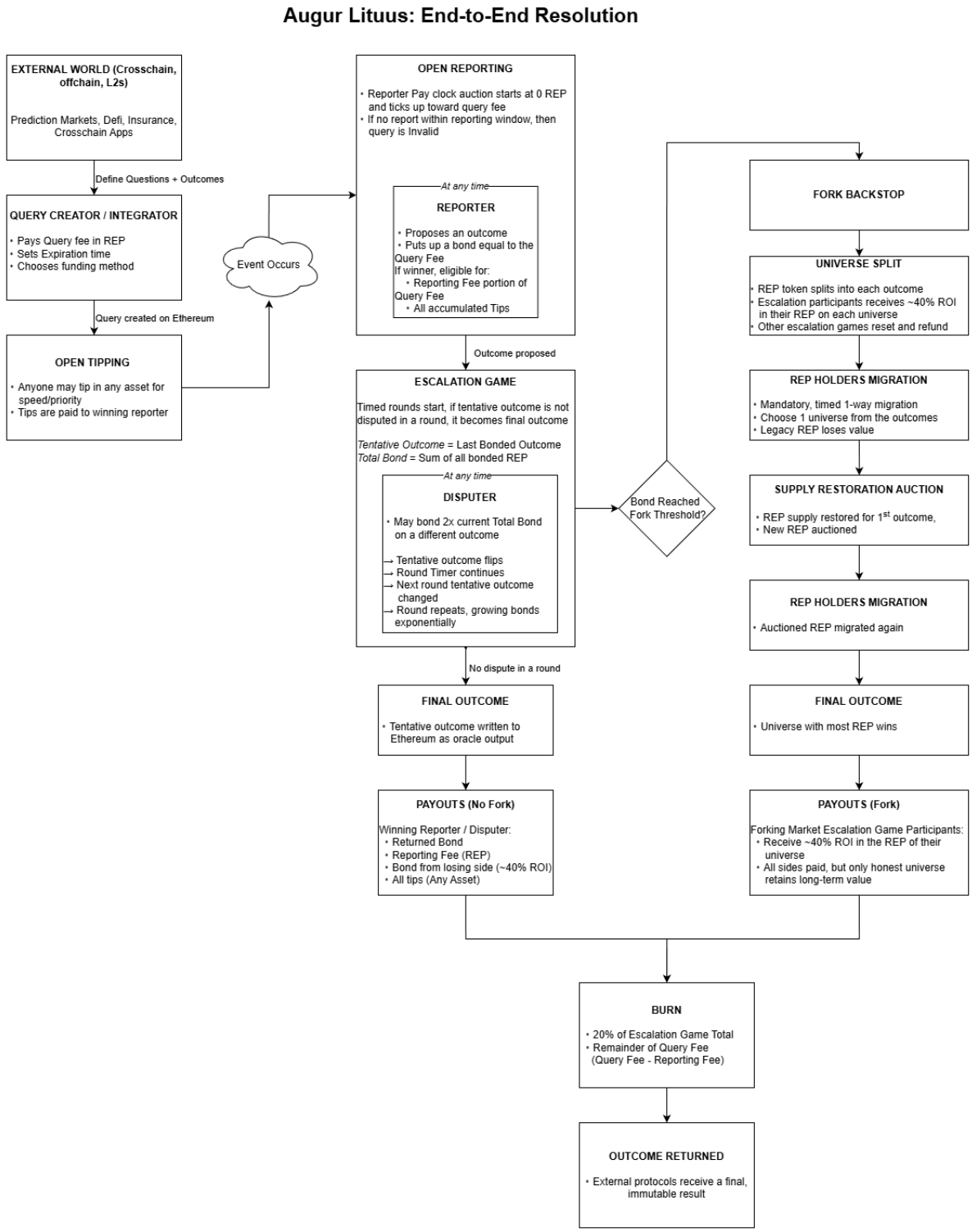
Appendix

Appendix.1 Specification Diagram



Appendix.2 Ground Truth Alignment

We argue that universe-forking oracle protocols align to ground truth better than any centralized entity. In an environment where future customers collectively direct their revenue in alignment with ground truth, the best strategy for every shareholder is to convert their shares to the fork representing ground truth. This holds true even if the majority of shareholders are tricked by a popular falsehood; when the majority migrates incorrectly, the majority will lose money, and the minority will profit. This stands in contrast to Shelling-point type voting systems where the incentive is to be in the majority rather than to be correct.

To be technically precise, in a market where users pay for access to a truth-as-a-service provider (i.e. a system that issues judgments on factual claims about reality), a mechanism that duplicates the provider by creating two identical versions differing only in a claim about reality where every shareholder is required to convert shares to a single version, will, incentivized by their valuation of the resulting assets, select the judgment that approximates the integrated knowledge of humanity more accurately than any centralized entity achieves alone.

This conclusion holds under the assumptions of rational market participants, a majority of future customers discerning ground truth, network effects in user adoption, the validity of the wisdom of the crowds hypothesis, and sufficient diversity and independence. We will first review each of these assumptions and then conclude by showing how they combine to induce participants to align to ground truth to the best of their ability. Under the wisdom of the crowds hypothesis, this aggregate attempt to align to ground truth provides answers more accurately than any single person.

Appendix.2.1 WoC hypothesis

The Wisdom of the Crowds hypothesis introduced by James Surowiecki[11] argues the aggregated judgment of a diverse group of independent individuals usually surpasses the accuracy of any single member's estimate, given sufficient diversity, independence, and effective aggregation. The validity of this hypothesis is supported by both theoretical foundations and empirical evidence across domains.

Theoretically, it leverages the law of large numbers and the diversity prediction theorem, which states that collective error equals average individual error minus prediction diversity; thus, greater diversity reduces error.

Empirically, studies demonstrate its efficacy in estimation tasks. An early example was the 1906 world fair, where 787 participants guessed the dressed weight of an Ox. The average of answers was 0.02% inaccurate and despite the fact that multiple butchers participated in the competition, the second best guess was 0.17% inaccurate. Note that the original study had mathematical errors, so the 2014 correction is referenced instead.[15]

Another study conducted at a 2015 TEDx event had audience members answer eight trivia questions with numerical answers such as "What is the height of the Eiffel Tower?" The average of all initial reports in the auditorium (N=5180) led to 52% error reduction compared to individual estimates. They also found that such error is further reduced by debate. When as little as four groups of five people debate and average together their 4 reports, they are able to outperform the average report of 1,400+ individuals.[8].

A 2022 study of 497 analysts rating 1275 upcoming cryptocurrency ICOs from January 2016 through December 2018 found that when the past-performance-weighted analyst rating increased by 1 standard deviation, the chance of raising enough money to meet the soft cap increased by 14 percentage points. Furthermore, they found a positive correlation (0.042) between accuracy and the number of analysts yielding another example of a crowd outperforming an individual.[6]

The Wisdom of the Crowds was also found to apply to earnings forecasting. On the Estimize platform 2516 members, only a third of whom are expert analysts, made 38k forecasts of earnings for 730 firms. The crowd had an average absolute error of 8 cents, outperforming Wall Street experts from IBES who have an average error of 9 cents. They also found evidence suggesting that people in the crowd who are further from correct actually help the crowd on average. Specifically, people who view the consensus before reporting became more accurate individually but caused the crowd's average to get worse 59% of the time.[3] Relevant to the context of this paper, this study suggests a crowd can predict better than experts which oracle outcome will generate the most profit.

Critics of WoC commonly highlight instances where it fails due to insufficient diversity or correlated errors. These failures can occur in two ways within our context. The first is when future customers of

the oracle tend to have an ideology or perception that differs from broader humanity. If this happens, the equilibrium would be to give future customers what they want even if it is less likely to be ground truth. Secondly, the majority of shareholders might migrate to a fork that receives less revenue because their lack of diversity led to a correlated error. If this happens, the oracle outcome is less likely to be ground truth, but nevertheless, those who converted their shares to the minority fork will receive the greatest payoff.

These exceptions underscore the importance of the independence and diversity conditions rather than invalidating the hypothesis. Overall, when conditions are met, WoC reliably produces superior outcomes, as evidenced by reduced variance and higher accuracy in controlled tests.

Appendix.2.2 Truth Preference Assumption

Customers prefer a provider who has historically issued judgments that they believe are ground truth. This preference exists because the customer has contracts with payoffs that depend on the provider's judgments. Unexpected judgments create variability in payoffs, which reduces utility for risk-averse customers, as their utility function is concave with respect to the payoff.

Appendix.2.3 Correct Majority Assumption

A majority of future oracle customers can discern ground truth. This assumption breaks when the majority of future customers make a false conclusion about reality, in which case the oracle will align to what future customers want rather than to ground truth. Even when this assumption breaks, three more criteria are required to violate the main claim that ground truth is more likely to come from a universe-forking oracle than a centralized oracle. First, the oracle must be given a question where a centralized entity is more likely to be correct than a majority of humanity. Second, the centralized entity must irrationally choose to reduce their own profits by issuing the correct answer. Third, the aforementioned type of question where an altruistic entity outperforms the crowd must make up a large portion of questions asked; large enough to overcome the accuracy increase on other questions from incentivizing the market to discover truth.

Appendix.2.4 Network Preference Assumption

Customers prefer a provider that has more customers over one that has less. This preference exists for two reasons. First, because the counterparty for the dependent contracts is more likely to accept the more popular provider, reducing the effort and/or chance to find a counterparty thereby increasing utility. Second, because the popularity of a provider is correlated with security through the following chain: increased popularity implies increased profit, which implies increased fully diluted valuation which is proportional to the cost to induce fraud, which is inversely correlated to the chance the dependent contract will experience losses due to fraud. Thus, a more popular provider is less likely to issue fraudulent outputs, increasing utility.

Appendix.2.5 Diversity Assumption

The shareholders, future oracle customers, and the broader market have diverse perspectives, knowledge, and/or approaches to problem-solving sufficiently similar to the set of all possible observers. In essence, this is assuming participants are a random subset of humanity. In this context, "sufficiently" means a degree of correlation that is enough to satisfy the WoC hypothesis.

Appendix.2.6 Overall Preferences

Considering the Truth Preference Assumption in the absence of the Network Preference Assumption, customers would always choose the provider that historically has best aligned with their personal assessment of ground truth. However, when these preferences combine, the overall preference is biased towards the more popular provider. For example, if 20% of customers believe B is true, some amount less than 20% (e.g. 5% or 0%) will use a provider that claims B is true due to network effects.

The incentives to align to ground truth back-propagate through the mechanisms of the protocol in many steps: Initially, we combine the Correct Majority and Truth Preference assumptions to conclude a majority of future customers prefer ground truth. This preference dictates future revenue flows between two oracles

that are identical except for a single ground truth claim. Next, the market uses the WoC hypothesis to accurately predict future revenue, the majority of which flows towards ground truth, thereby setting a higher market valuation for the fork representing ground truth. Next, diverse shareholders subject to the migration mechanism of the forking game both predict and migrate proportional to the market valuation by predicting ground truth accurately per the WoC hypothesis. Finally, at the outset, individual reporters are incentivized to the best of their ability to report ground truth knowing that deviation will trigger profitable escalation to the forking game where they will lose their reporting bond because as previously established the forking game aligns to ground truth. Furthermore, the Network Preference Assumption creates a bias that further aligns the outcome closer to ground truth.

Note that this chain of consequences is most weak at the point that critically relies on shareholders being diverse within the context of the WoC hypothesis. This group is the smallest and least diverse, and thus will predict ground truth less accurately than other groups like the broader market, leading to a possible error. When such an error occurs, those who migrated in conflict with ground truth will lose money and the oracle will align to an outcome that is arguably true but further from ground truth than the alternative.

Overall, under these assumptions, we conclude Universe-Forking to align more accurately to ground truth than a centralized entity.

Appendix.3 Derivation: SP

The securityMargin formula is:

$$\text{securityMargin} = \text{attackCost}/\text{naturalFDV}$$

An attacker of Schelling-point oracles makes an investment of REP that gives majority control, votes maliciously, and then sells the shares at the attacked valuation. The net attack cost then is given by:

$$\text{attackCost} = \text{attackInvestment} \times (1 - \text{attackedValuation})$$

This modeling does not include hedging but this can be considered by increasing the assumption of attacked valuation. If the attack investment is fully hedged, the attack cost approaches zero as friction is removed, if the attack investment is 50% hedged, the attack valuation should be increased from the predicted value halfway to 1.

The cost to purchase half of the responsive supply quantified as attackInvestment primarily depends on price impact. A large purchase will increase the price of an asset. This holds even when purchased slowly to allow time for those without limit orders to notice that the price meets their reservation price. Without this premium, buying half of the supply would cost half of the FDV. Additionally, less than half of the supply is required if every shareholder does not vote, which reduces the 50% requirement by responsiveREP. Consequently, the formula for attack investment is 50% of responsiveREP * FDV * premium:

$$\text{attackInvestment} = 0.5 \times \text{responsiveREP} \times \text{naturalFDV} \times \text{premium}$$

The formula for Schelling-point voting security margin then is:

$$\text{securityMargin(SP)} = 0.5 \times \text{responsiveREP} \times \text{premium} \times (1 - \text{attackedValuation})$$

Appendix.4 Derivation: PBUF

The securityMargin formula is:

$$\text{securityMargin} = \text{attackCost}/\text{naturalFDV}$$

In Price-Based Universe Forking (PBUF), every share is split into a share from every fork. The fork with the greatest market price wins. An attacker of PBUF must offer to buy the lesser-value asset, termed false-REP, at the same price as the greater-value asset. This expense represents an attack investment, and

the net attack cost is this attack investment minus what the attacker is able to sell the false-REP shares for after the attack:

$$\text{attackCost} = \text{attackInvestment} - \text{falseProceeds}$$

This offer to buy false-REP above its market value makes it rational for all responsive shareholders to sell their false-REP shares. The buyers can see the price manipulation which is only rational if an attacker has enough funds to win, giving buyers the expectation the attack will succeed. Consequently, they are only willing to buy at the post-attack valuation despite the attack being in progress. The attack investment then is the market share of the true fork times the natural FDV adjusted for the attacked valuation times the portion of shareholders that are responsive to incentives:

$$\text{attackInvestment} = (1 - \text{falseMarketShare}) \times \text{naturalFDV} \times \text{attackedValuation} \times \text{responsiveREP}$$

The attacker recovers proceeds from selling the false supply they purchased during the attack. The portion they purchased was the responsive REP and it sells for the false portion of the natural FDV with a reduction in valuation due to an attack:

$$\text{falseProceeds} = \text{responsiveREP} \times \text{falseMarketShare} \times \text{naturalFDV} \times \text{attackedValuation}$$

Combining the above formulas yields the security margin formula for Price-Based Universe-Forking:

$$\text{securityMargin(PBUF)} = \text{responsiveREP} \times \text{attackedValuation} \times (1 - 2 \times \text{falseMarketShare})$$

Appendix.5 Derivation: MBUF

The securityMargin formula is:

$$\text{securityMargin} = \text{attackCost}/\text{naturalFDV}$$

An attacker of Migration-Based Universe-Forking must purchase half of the non-sleeping supply and convert it into a fraudulent fork of REP. Additionally, as will be explained later, the existence of unresponsive REP will increase the entry price. Thus, the money they lose in-protocol is the attack investment adjusted for unresponsiveREP minus any money they recover from selling false-REP as represented by:

$$\text{attackCost} = \text{attackInvestment} \times \text{unresponsivePriceAdjustment} - \text{falseProceeds}$$

Under the strongest form of this protocol, the migration rule destroys all REP that does not choose a fork. This destruction increases the portion of the supply that each share represents. The market estimates the portion that will not respond and prices this in, knowing that if anyone triggers a fork the gains will be realized. If the market did not price it in, the price per share would increase when a fork happens due to a supply reduction occurring, while the diluted valuation remains the same. Put another way, before a fork the market behaves as if a portion of the apparent supply does not exist; so what on paper is a 1% ownership is actually greater than 1% in expectation and thus will sell for more per share, artificially increasing the FDV and attack cost. The price adjustment for unresponsive REP is given by:

$$\text{unresponsivePriceAdjustment} = 1/\text{responsiveREP}$$

The Migration-based attack investment is the same as Schelling-point because both mechanisms require buying half of the responsive supply at a premium to change the oracle output:

$$\text{attackInvestment} = 0.5 \times \text{responsiveREP} \times \text{naturalFDV} \times \text{premium}$$

The 'false proceeds' is the money the attacker might recover by selling false-REP. The proceeds are the natural attacked valuation of false-REP times the portion the attacker owns represented by the final term in the formula below. For example, if false-REP is worth slightly less than true-REP, the attacker only needs to migrate a slight amount with the remainder rationally migrating. The formula is:

$$\text{falseProceeds} = \text{naturalFDV} \times \text{falseMarketShare} \times \text{attackedValuation} \times (1 - 2 \times \text{falseMarketShare})$$

Combining the above formulas yields the security margin formula for Migration-Based Universe Forking:

$$\text{securityMargin(MBUF)} = 0.5 \times \text{premium} - \text{falseMarketShare} \times \text{attackedValuation} \times (1 - 2 \times \text{falseMarketShare})$$

Note that unlike Augur V2, other MBUF protocols such as Kleros V2 and Eigenlayer do not destroy unresponsive shares during a fork. This lack of supply reduction eliminates the unresponsivePriceAdjustment, thereby preventing responsiveREP from canceling out, reducing the security margin to:

$$\text{securityMargin(MBUF without mandatory migration)} = \text{responsiveREP} \times \text{securityMargin(MBUF)}$$

Appendix.6 Derivation: MBUFSR

The securityMargin formula is:

$$\text{securityMargin} = \text{attackCost}/\text{naturalFDV}$$

An attacker of a Migration-Based Universe-Forking with Supply Restoration oracle must make investments in REP, then convert that REP into the less valuable fork. The attacker then recovers a portion of their investment in the event false-REP has natural value. The in-protocol loss experienced by an attacker is:

$$\text{attackCost} = \text{attackInvestment} - \text{falseProceeds}$$

The required attack investment includes both an initial investment and an investment of the minted supply. The initial investment is a majority of the responsive REP less the portion that rationally migrates to false. This portion that rationally migrates to false is equal to the false-REP market share. After adjusting for the natural FDV and the premium on the initial investment, the capital required to conduct an attack is:

$$\text{attackInvestment} = (0.5 - \text{falseMarketShare}) \times \text{responsiveREP} \times \text{naturalFDV} \times \text{premium} + \text{mintedInvestment}$$

In the unlikely event the market expects false-REP to earn legitimate profit, the attacker obtains proceeds from selling false-REP after an attack. The proceeds are the portion of the false supply the attacker owns (attackersPortionOfFalse) times the FDV of a post-attack false-REP. This FDV includes the discounted portion of the subsidy that will go to false-REP (explained later). This formula is:

$$\begin{aligned}\text{falseProceeds} = \text{attackersPortionOfFalse} \times \text{falseMarketShare} \times \\ (\text{naturalFDV} \times \text{attackedValuation} + \text{discountedSubsidy})\end{aligned}$$

The amount the attacker must spend purchasing supply that was minted during the supply restoration is:

$$\text{mintedInvestment} = \text{mintedInvestmentInitial} + \text{mintedInvestmentAdditional}$$

The attacker's initial investment of minted shares is the minting price times half of the quantity initially minted:

$$\text{mintedInvestmentInitial} = \text{mintPrice} \times 0.5 \times \text{initialMintQuantity}$$

The initial quantity minted is the amount needed to restore the supply. This is the reduction due to unresponsive "sleeping" REP being destroyed plus the attackers migration to false before the minting phase.

$$\text{mintQuantityInitial} = \text{supply} \times ((1 - \text{responsiveREP}) + \text{responsiveREP}/2)$$

After the attacker migrates half of the initially minted shares, the supply falls short of a full restoration caused by and proportional to the amount migrated to false-REP. The amount migrated to false-REP is always half because this is the amount required for the attacker to win. As such, in the second restoration half as much is minted, and in the third restoration a fourth as much is minted. This series $(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots)$ equals 1 in the limit. Thus, the additional minted investment is equal to the initial minted investment:

$$\text{mintedInvestmentAdditional} = \text{mintedInvestmentInitial}$$

It also follows that the total mint quantity is double the initial mint quantity. Simplified this is:

$$\text{mintQuantity} = \text{supply} \times (2 - \text{responsiveREP})$$

The minting price is the natural FDV of true-REP expressed by naturalFDV $\times$ (1 $-$ falseMarketShare) divided by expected supply which is the same as the starting supply due to restoration. However, the minting price is reduced by the inefficiency of the auction and the attacked valuation, and it is increased by the impact of the subsidy on true-REP's FDV expressed by discountedSubsidy * (1 - falseMarketShare). Finally, it is divided by supply to yield a unit minting price of:

$$\text{mintPrice} = \frac{\begin{array}{l}\text{naturalFDV} \times (1 - \text{falseMarketShare}) \times \text{auctionEfficiency} \times \text{attackedValuation} + \\ \text{discountedSubsidy} \times (1 - \text{falseMarketShare})\end{array}}{\text{supply}}$$

After an attack where the false market share is zero, the attacker is the only one to migrate to false meaning 100% of the false-REP supply will be in the possession of the attacker; this is expected if the winning outcome is obvious fraud. However, if the Query is legitimately contentious, false-REP may expect revenue, leading to the same portion of the REP supply rationally migrating to false. This rational migration reduces the portion of the false supply that the attacker owns after an attack. The portion of false-REP owned by the attacker after all supply restorations and migrations is:

$$\text{attackersPortionOfFalse} = 1 - \text{falseMarketShare}$$

During supply restoration, the protocol auctions REP in exchange for a different asset. These auction proceeds are a side effect of the attackers actions and could be destroyed without harming any participant; however, to improve the security margin this asset is used to subsidize legitimate use by refunding blockchain costs for all forks of the protocol. This subsidy can easily take years to spend and is thus time-discounted when accounting for its ability to increase the FDV. The subsidy discount is normalized to when the subsidy equals natural FDV, so the following formula scales this discount proportional to the subsidy divided by naturalFDV. Thus, the amount the subsidy increases the cumulative FDV of all forks is:

$$\text{discountedSubsidy} = \frac{\text{subsidy} \times \text{subsidyDiscount}}{\text{subsidyDiscount} + (1 - \text{subsidyDiscount}) \times \left(\frac{\text{subsidy}}{\text{naturalFDV}}\right)}$$

The subsidy is the proceeds from the supply restoration which is the total quantity minted times the minting price:

$$\text{subsidy} = \text{mintQuantity} \times \text{mintPrice}$$

Substituting the above relationships results in a recursive solution because the discounted subsidy affects its own size. As a result, the simplified form is a fairly complicated closed-form solution to this transcendental relationship, yielding the following formula for the security margin of Migration-Based Universe-Forking with Supply Restoration:

$$\text{securityMargin(MBUFSR)} =$$

$$\begin{aligned}&(0.5 - \text{falseMarketShare})\,\text{responsiveREP premium} \\ &\quad + \left(\text{auctionEfficiency}(1 - \tfrac{\text{responsiveREP}}{2}) - \text{falseMarketShare}\right) \\ &\quad \times \left(\frac{-\begin{bmatrix}1 - (1 - \text{falseMarketShare})\text{attackedValuation}\left(\frac{1}{\text{subsidyDiscount}} - 1\right)(2 - \text{responsiveREP})\text{auctionEfficiency} \\ -(1 - \text{falseMarketShare})(2 - \text{responsiveREP})\text{auctionEfficiency}\end{bmatrix} + \text{root}}{2\left(\frac{1}{\text{subsidyDiscount}} - 1\right)(2 - \text{responsiveREP})\text{auctionEfficiency}}\right),\end{aligned}$$

where

$$\text{root} = \sqrt{\begin{array}{l}\begin{bmatrix}1 - (1 - \text{falseMarketShare})\text{attackedValuation}\left(\frac{1}{\text{subsidyDiscount}} - 1\right)(2 - \text{responsiveREP})\text{auctionEfficiency} \\ -(1 - \text{falseMarketShare})(2 - \text{responsiveREP})\text{auctionEfficiency}\end{bmatrix}^2 \\ + 4\left(\frac{1}{\text{subsidyDiscount}} - 1\right)(2 - \text{responsiveREP})\text{auctionEfficiency}(1 - \text{falseMarketShare})\text{attackedValuation}\end{array}}.$$

Appendix.7 Conjecture: Plausible Scenarios

The security margin of each oracle protocol is sensitive to various assumptions regarding the external state. As such, the methodology of this paper is to compare security margins under a range of plausible conditions. This subsection has conjecture regarding plausible ranges of each input variable to the security margin formulas.

	Possible Range	Plausible Range	Most Likely
attackedValuation	$[0, 1]$	$[0.5, 1]$	0.7
subsidyDiscount	$(0, 1]$	$[0.08, 0.94]$	0.52
responsiveREP	$[0, 1]$	$[0.7, 1]$	0.9
auctionEfficiency	$(0, 1)$	$[0.7, 0.99]$	0.9
premium	$[1, \infty)$	$[1.07, 5.3]$	1.9
falseMarketShare	$[0, 0.5)$	$[0, 0.15]$	0.04

Appendix.7.1 attackedValuation

The combined natural value of all forks after an attack divided by the starting natural FDV. The possible range is [0, 1]. It is likely on the high side of this range because a single instance of fraud can not be expected to eliminate all future customers. Theoretically, an attack has no impact on revenue because an attack is a consequence of the vulnerable open interest exceeding the security margin, a circumstance resulting from irrationally betting excessive amounts on outcomes rather than a flaw of the protocol design. An attack further incentivizes future customers to limit their collective betting to less than the security margin, but in theory does not decrease demand beyond achieving this objective. However, potential customers are not likely to see it this way.

The plausible range of attackedValuation differs between Schelling-point and Universe-Forking protocols. An attack on a universe forking protocol should theoretically result in little to no reduction of revenue because the attackers shares are converted to a different fork where they can not harm future customers. However, in practice, the protocol will still likely suffer a loss of revenue due to fear and bad press. In a Schelling-point oracle, on the other hand, the attacker still has majority control of the voting process after an attack enabling future attacks, thereby deterring customers. The attacked valuation for an SP oracle without a credible threat of social recovery is arguably lower than it is for other oracle protocols. Also note that in both cases an attacker is expected to hedge by shorting REP, an act that is not part of this model, but one that effectively increases the attacked valuation. As such, we believe the plausible range is [0.5,1) and the most likely value is 0.7. This 0.7 could be seen as expecting a 30% loss of revenue or, for example, a 60% loss of revenue with the attacker half hedged.

Appendix.7.2 subsidyDiscount

This is the portion of the subsidy that translates into increased FDV due to rebates on blockchain transaction costs for legitimate use. This is normalized to when naturalFDV=subsidy. For example, if subsidyDiscount is 52%, then when the subsidy is equal to the natural FDV, the increase to the FDV caused by the addition of a subsidy is 52%. The possible range is (0, 1]. To find the plausible range, we modeled the yearly blockchain transaction costs of competing and similar protocols relative to their FDV. For example, Augur V2 was at the high end at 4% of FDV spent on blockchain costs per year, Uniswap and UMA are 0.8%, and Polymarket was the lowest at 0.006%, which normalizes to 1.1% if Polymarket had deployed on Ethereum L1 instead of Polygon. We concluded that the plausible range was 0. 4% to 4% per year and the likely value was 1%. From this starting point, time discounting was applied. Time discounting can arguably be 0% if the airdrop is a yielding asset, is likely 1.5% based on recent economic data, and can come with a premium in a blockchain setting, increasing the discounting as high as 5% per year. Using an expense rate of 0.4% to 4% per year for a subsidy equal to the FDV, and a time discounting of 0.5% to 5% per year, yields, via the average value of an exponential decay, a plausible range for subsidyDiscount of 8% to 94% with a likely value of 52%.

Appendix.7.3 responsiveREP

The portion of REP that will respond to incentives. Theoretically, this value should be 1 because it is irrational to not respond to incentives. Prior work generally assumes 100% of participants are rational.

Some protocols quickly diminish in robustness as irrationality increases, and so for real world modeling this factor is important to include. For Schelling-point oracles, this is the portion of holders that will participate in a governance vote due to likely loss of share value if the consensus is fraudulent. For universe-forking oracles, this is the portion of REP that will migrate to a fork incentivized by the total loss of their shares if they fail to do so. In either case the shareholder has a strong incentive to participate and is generally given weeks of time to become informed. As such, we find the plausible range to be 70% to 100% with a likely value of 90%.

Appendix.7.4 auctionEfficiency

Percent of the fair price that REP will sell for when time sensitive (e.g. auction). Auctions for large portions of the supply within weeks likely exceed the capacity of typical buyers. This incentivizes arbitrage to buy below fair value and wait to resell closer to fair value. Exposure to REP introduces risk and thus arbitrage may demand a significant discount. The arbitrage returns increase hyperbolically as auctionEfficiency decreases, while at the same time the risk of the trade being unprofitable decreases. As such, we assess a plausible range of 70% to 99%, and a likely value of 90%.

Appendix.7.5 premium

The average cost paid by the attacker to purchase REP before attacking divided by the starting price. This includes trading fees and opportunity cost. We expect that attempting to slowly buy half of the supply will result in an average purchase price inclusive of opportunity cost and fees that is at least 7% more than the starting price, likely 90% more, and at most 430% more. This low end is only possible in a liquid market that is able to robustly short overvaluation, while the likely and maximum value were obtained via modeling of reservation prices under a lognormal distribution with our predictions of what price increase would be required for half of the holders to sell.

Appendix.7.6 falseMarketShare

The market share of the second most valuable fork. For example, if true-REP expected 80% of revenue and false-REP 20%, the falseMarketShare would be 0.2. Prior works generally assume a falseMarketShare of 0%. This is because for nearly all queries, the correct answer is obvious, meaning false-REP is obvious fraud and will expect no market share. However, a falseMarketShare reduces the attack cost, and thus we can expect the attacker to fork or even create the most legitimately contentious Query possible. A "legitimately contentious" Query is one in which two perspectives each see a different clear answer and both sides agree it should not be contentious (otherwise they would agree to resolve as 'invalid'). Rationally, a proportion of REP will migrate to false equal to the false market share, reducing the attack cost. However, in practice, even if say 20% of customers prefer to use an oracle that reported a specific outcome, due to network effects only a small fraction of these will be bothered enough to use the ecosystem with less liquidity and counterparties. Put simply, even for legitimately contentious issues, we expect virtually all customers to use the fork with the greater natural market value even if they disagree with a particular ruling. As a case study consider the legitimately contentious fork that created Ethereum Classic which received only a few percent market share after forking despite a far more than a few percent of users preferring the "code is law" approach to "social recovery". As such we find the plausible range for falseMarketShare to be 0% to 15%, with a likely value of 4%.

Appendix.8 Proof: MBUF$\geq$ SP

To prove this claim, consider the simplified difference MBUF - SP:

$$0.5 \times \text{premium} \times (1 - \text{responsiveREP}) + \text{attackedValuation} \times$$
$$(0.5 \times \text{premium} \times \text{responsiveREP} + 2 \times \text{falseMarketShare}^2 - \text{falseMarketShare})$$

with attackedValuation $\in [0, 1]$, responsiveREP $\in [0, 1]$, premium $\in [1, \infty)$, and falseMarketShare $\in [0, 0.5)$. To demonstrate that MBUF $\geq$ SP, we establish a lower bound for each of the terms to show the expression always resolves to greater than or equal to zero.

The first term is:

$$0.5 \times \text{premium} \times (1 - \text{responsiveREP}) \geq 0$$

with equality if and only if responsiveREP = 1. The second term is:

$$\text{attackedValuation} \times (0.5 \times \text{premium} \times \text{responsiveREP} + 2 \times \text{falseMarketShare}^2 - \text{falseMarketShare})$$

This second term is attackedValuation times a bracket. If attackedValuation were 0, the second term would resolve to 0 satisfying the inequality. Thus the capacity for negative numbers is constrained to this bracket evaluating to a negative number. The least possible value for the bracket is achieved by setting $0.5 \times \text{premium} \times \text{responsiveREP}$ to 0 which requires responsiveREP to equal 0, and by setting the remaining terms as low as possible. The remaining terms are $2 \times \text{falseMarketShare}^2 - \text{falseMarketShare}$. This is a quadratic function in falseMarketShare, which is a parabola opening upwards. Its vertex (minimum point) occurs at falseMarketShare = 0.25, which is within $[0, 0.5)$. Substituting this minimizing value into the remaining terms gives -0.125.

The overall minimum depends on cases based on the first term.

The first case is responsiveREP = 1. Here, the first term becomes zero, and the second term added to it must be non-negative because the first part of the bracket is at least 0.5 and as previously established the remaining terms are at least -0.125. This first case is positive because 0.5-0.125>0.

The second case is when responsiveREP<1 resulting in a positive first term and a possibly negative second term added to it. Consider how the overall value changes with respect to responsiveREP. The first term decreases at a rate of $-0.5 \times \text{premium}$ as responsiveREP increases. The second term increases at a rate of attackedValuation $\times 0.5 \times$ premium as responsiveREP increases. Thus, as responsiveREP changes from 1 to lesser values, the overall value for the difference increases from its already positive value (or remains constant when attackedValuation = 1).

In conclusion, MBUF=SP only if responsiveREP=1 and attackedValuation=0. Outside of this boundary condition, MBUF>SP because the first term is positive and the second term added to it can never be more negative than the first term is positive.

Appendix.9 Proof: MBUFSR>SP|A

Under the set of scenarios A defined by complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), MBUFSR is strictly more secure than SP. To prove this claim, consider the simplified difference MBUFSR-SP under these scenarios:

$$\frac{\text{premium}}{2} + \frac{\text{auctionEfficiency} \times \text{subsidyDiscount}}{\text{subsidyDiscount} - \text{auctionEfficiency} + \text{root}},$$

where $\text{root} = \sqrt{(\text{auctionEfficiency} + \text{subsidyDiscount})^2 - 4 \times \text{auctionEfficiency} \times \text{subsidyDiscount}^2}$

We will now show this expression is always greater than 0 and consequently that MBUFSR always has a greater securityMargin than SP within this set of scenarios. The first term is minimized to 0.5 when 'premium' is at its least possible value 1. So we have a positive term plus a fraction. Next we show this fraction is always positive. The numerator is the product of two positive numbers, a value that must be positive. The possibility of a negative outcome then is constrained to a negative denominator. To show the denominator is always positive, we rewrite it as:

$$\text{subsidyDiscount} - \text{auctionEfficiency} + \text{root}$$

where:

$$\text{root} = \sqrt{(\text{auctionEfficiency} + \text{subsidyDiscount})^2 - 4 \times \text{auctionEfficiency} \times \text{subsidyDiscount}^2}$$

The square root function always yields a non-negative value. Thus, root is positive. To establish that the denominator is greater than 0, two cases must be considered.

In the first case where subsidyDiscount is at least auctionEfficiency, they subtract to a non-negative number. At the boundary of this case:

$$\text{subsidyDiscount} = \text{auctionEfficiency} < 1,$$

then root>0 ensuring denominator>0. Thus in the first case, the denominator and every other part of the expression are greater than zero.

In the second case where subsidyDiscount is less than auctionEfficiency, in order to show the whole denominator>0, we need to prove that the positive contribution from root more than offsets the negative contribution of subsidyDiscount - auctionEfficiency. For this we re-arrange 'root':

$$\text{root} = \sqrt{(\text{auctionEfficiency} - \text{subsidyDiscount})^2 + 4 \times \text{auctionEfficiency} \times \text{subsidyDiscount} \times (1 - \text{subsidyDiscount})}$$

where the first term of 'root' is a square and therefore always positive and the second term of 'root' is also positive because it is only positive numbers multiplied together. These terms are added together, so the inside of the root is at least the value of the first term. Since we are in case 2, the second term of root must be greater than zero. Thus, 'root' is at least:

$$\text{auctionEfficiency} - \text{subsidyDiscount}$$

plus some positive value termed positiveNumber. Substituting this lower bound into the denominator gives:

$$\text{subsidyDiscount} - \text{auctionEfficiency} + (\text{auctionEfficiency} - \text{subsidyDiscount}) + \text{positiveNumber}$$

which simplifies to 'positiveNumber'. Thus in all possible cases of complete information and rationality, MBUFSR strictly has more security margin than SP.

Appendix.10 Proof: MBUFSR>SP|B

Under the set of scenarios B defined by plausible values: attackedValuation $\in [0.5, 1]$, subsidyDiscount $\in [0.08, 0.94]$, responsiveREP $\in [0.7, 1]$, auctionEfficiency $\in [0.7, 0.99]$, premium $\in [1.07, 5.3]$, falseMarketShare $\in [0, 0.15]$, MBUFSR is strictly greater than SP. To prove this claim, let a = attackedValuation, d = subsidyDiscount, r = responsiveREP, e = auctionEfficiency, p = premium, f = falseMarketShare and consider the simplified difference MBUFSR − SP:

$$\frac{2epr(a-2f)(d-1)(r-2) + (e(r-2)+2f)(ae(d-1)(f-1)(r-2) + d(-e(f-1)(r-2) - \text{root} + 1))}{4e(d-1)(r-2)}$$

where

$$\text{root} = \sqrt{\frac{-4ade(d-1)(f-1)(r-2) + (ae(1-d)(1-f)(2-r) + d(e(1-f)(2-r)-1))^2}{d^2}}$$

The minimum occurs at attackedValuation=0.5, subsidyDiscount=0.08, responsiveREP=1, auctionEfficiency=0.7, premium=1.07, falseMarketShare=0.15. The expression is continuous, so this minimum can be verified as we have by making small perturbations to each of these values and observing that the difference increased. At this minimum, the difference evaluates to $0.204 > 0$, and is therefore strictly positive over the ranges.

Appendix.11 Proof: MBUFSR>MBUF|A

Under the set of scenarios A defined by complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1), MBUFSR is strictly more secure than MBUF. To prove this claim, consider the simplified difference MBUFSR-MBUF under these scenarios:

$$\frac{\text{auctionEfficiency}}{(1 - \text{auctionEfficiency}/\text{subsidyDiscount}) + \text{root}}$$

where

$$\text{root} = \sqrt{(1 - \text{auctionEfficiency}/\text{subsidyDiscount})^2 + 4 \times \text{auctionEfficiency} \times (1/\text{subsidyDiscount} - 1)}$$

To show that both the numerator and denominator are positive and thus the difference is positive, recall subsidyDiscount $\in (0, 1]$ and auctionEfficiency $\in (0, 1)$, so the numerator is greater than 0. The denominator can be represented as:

$$(1 - b) + \text{root}$$

where:

$$b = \text{auctionEfficiency}/\text{subsidyDiscount}$$

$$\text{root} = \sqrt{(1-b)^2 + 4 \times \text{auctionEfficiency} \times (1/\text{subsidyDiscount} - 1)}$$

Observe that b is positive because it is a positive number divided by another. To show the denominator is positive, consider two cases. In the first case where $b \leq 1$ (i.e. auctionEfficiency $\leq$ subsidyDiscount), the expression is a positive term plus a square root. The square root of any number is positive, thus the first case is the addition of two positive numbers and is overall positive. In the second case where $b > 1$ the first term becomes negative, so we need to show that:

$$(1 - b) + \text{root} > 0$$

which rearranges to:

$$\text{root} > b - 1$$

Observe that both root and $b - 1$ are positive, so squaring both sides will not change the sign:

$$(1-b)^2 + 4 \times \text{auctionEfficiency} \times (1/\text{subsidyDiscount} - 1) > (b-1)^2$$

To check if $\text{root}^2 > (b-1)^2$, subtract the right from the left and substitute:

$$\left(1 - \frac{\text{auctionEfficiency}}{\text{subsidyDiscount}}\right)^2 + 4 \times \text{auctionEfficiency} \times \left(\frac{1}{\text{subsidyDiscount}} - 1\right) - \left(\frac{\text{auctionEfficiency}}{\text{subsidyDiscount}} - 1\right)^2 > 0$$

When the first and third terms are expanded they cancel out leaving only the second term as the simplified form:

$$4 \times \text{auctionEfficiency} \times (1/\text{subsidyDiscount} - 1) > 0$$

This is two positive values multiplied by $(1/\text{subsidyDiscount} - 1)$. As we are in case 2 defined by subsidyDiscount $<$ auctionEfficiency, subsidyDiscount must be less than 1. When 1 is divided by a lesser number, the result is greater than 1. So the final value is a number greater than 1 minus 1 which is positive. Thus, case 2 is all positive numbers multiplied together.

Overall we have shown that the numerator and both cases for the denominator can only be positive. Thus, MBUFSR is strictly more secure than MBUF conditional on complete information and rationality (falseMarketShare=0, responsiveREP=1, attackedValuation=1).